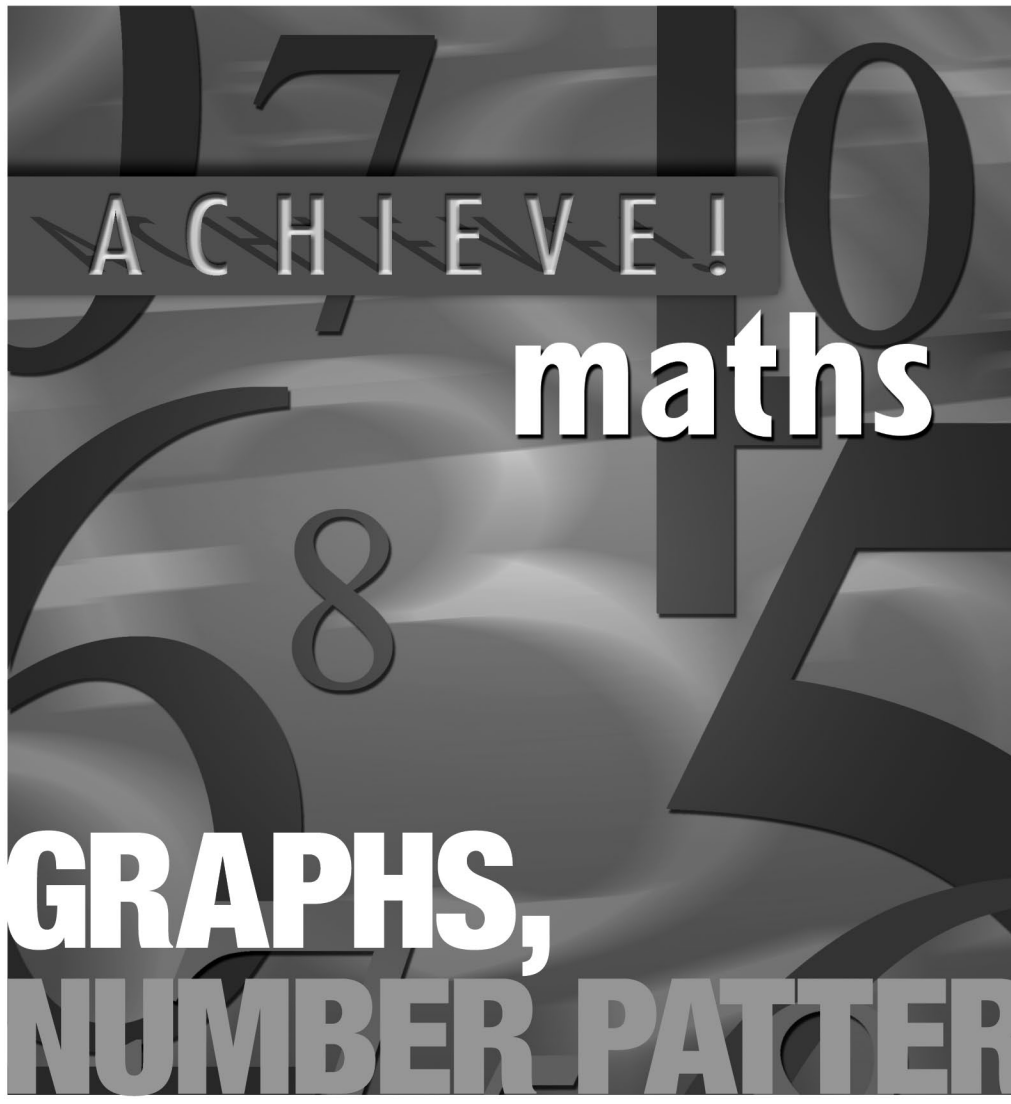
ACHIEVE!

maths

# GRAPHS, NUMBER PATTERNS & ALGEBRA

Paul Steenson




# GRAPHS, NUMBER PATTERNS & ALGEBRA

Paul Steenson

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and Gail McManus Graphics designed the cover.

**Paul Steenson** has more than thirty years' experience teaching Mathematics in secondary schools in Australia. He has also worked as a test development officer and been an assessor of Mathematics papers for the NSW HSC examination. Paul served as a member of the NSW Mathematics Syllabus Committee and has been a marker of both School Certificate and HSC examination papers. He has been a regular presenter of curriculum workshops and teacher development activities. He has also been a co-author of textbooks and written a number of titles in the Instant Lessons series.

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# Introduction

The *Achieve!* series of Maths titles has been developed for teachers who require modified, curriculum-focused activities for secondary school students. It is aimed at students who, because of poor literacy skills, struggle to read and comprehend regular classroom materials. Some activities are more challenging than others. The activity sheets are also suitable as supplementary material for regular classes. They may also provide useful revision in preparation for state-wide or national numeracy testing.

This book is divided into five units, each of which focuses on a topic or set of skills to do with graphs, number patterns or algebra. The CD contains all the pages that are featured in the book, as well as a file of illustrations that can be used for other activity sheets. The activity sheets are provided as interactive PDFs, which are ideal for use on interactive whiteboards (IWBs) and laptops. Teachers should note that students can save their answers and any other changes they make. For greater accuracy, questions that require students to measure, circle or draw should be completed on hard copies or on the IWB using the IWB tools.

Note that photocopying and printing copies from the CD may change the length of some lines and the size of diagrams. When students are asked to measure lengths, it is recommended that the teacher checks the answers using the same copy that is given to students.

Teachers can use the material in a variety of ways. For example, a unit could be taught in sequential lessons with a whole class or with students in groups of varying sizes. Alternatively, one or more activity sheets can be used as support material, for individual homework, assessment or revision. Teachers will need to select carefully if individual activity sheets are used as stand-alone tasks, as some activities build on knowledge gained earlier in the unit.

The Teacher's notes provide valuable guidance about using each unit, and contain the following elements.

***Objectives –*** These are the key skills and knowledge learnt through using the activity sheets.

***Prior knowledge –*** This refers to the skills and knowledge that are required for students to complete the tasks.

***Background –*** This section provides a range of different types of useful information to help teach this unit. It can include explanations of key concepts and terminology or further background notes about the subject matter.

***Starter activity –*** Warm-up activities are provided to put the content into proper context and to gain student interest and attention.

***Activity sheets and answers –*** A summary of the content of each activity sheet is provided, along with teaching tips and selected answers to assist with time-efficient marking.

***Extension ideas –*** Additional activities or discussion ideas are suggested to recap on the main skills, knowledge or terminology taught, as well as ways to extend, assess or consolidate student understanding.

***Curriculum links –*** The activity sheets will enable students to achieve outcomes required for the following State levels or stages:

- New South Wales – Stages 4 and 5
- Victoria – Levels 5 and 6
- Queensland – Levels 4, 5 and 6

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# Teacher's notes

## OBJECTIVES

- Recognise different types of statistical graph.
- Read and interpret data presented in various graphs.
- Recognise and interpret various scales used in graphs.
- Interpret a given line graph using the scales on the axes.

## PRIOR KNOWLEDGE

Students should able to perform the basic operations with whole numbers, fractions and decimals using mental arithmetic or a calculator.

## BACKGROUND

Column graphs may consist of vertical columns or horizontal bars. However, the term *bar graph* is only used with divided bar graphs. Avoid using this term for column graphs with horizontal bars (horizontal column graphs).

Sector graphs and divided bar graphs are used to show how a total is divided into parts.

## STARTER ACTIVITY

Show students a selection of different graphs and make sure there is more than one of each type. Ask students to name each type of graph and discuss the main features of each graph, such as heading or title, axes, scale and key. Then briefly discuss what each graph shows.

## ACTIVITY SHEETS

### 1.1 Car sales – a picture graph

In this activity sheet, students are given a simple picture graph, arranged vertically, in which each car symbol represents 1 car sold. They are asked to read and interpret data from this graph. In the last question they are asked to draw cars in the fifth column of the graph to show the number of cars sold in the 5th week.

**Answers**

**1** **a** 7 **b** 9 **2** 1st week **3** 4th week **4** 35

**5** 3

### 1.2 Canned fruit – a picture graph

In this activity sheet, students are given a picture graph, arranged horizontally, in which each symbol represents 10 cans of fruit sold. They are asked to read and interpret data from this graph.

**Answers**

**1** 10 **2** 5 **3** 70 **4** 85 **5** Pineapples

**6** Peaches **7** 15 **8** 320

### 1.3 Clean-up day results – a column graph

In this activity sheet, students are given a simple column graph. They are asked to read data from the graph and arrange it in a table.

**Answers**

**1** Paper **2** 30 kg **3** 45 kg

**4**

| Type of item | Mass collected |
|---|---|
| Aluminium cans | 30 kg |
| Steel cans | 35 kg |
| Plastic | 45 kg |
| Paper | 60 kg |
| Glass bottles | 40 kg |
| Other | 25 kg |

**5** 235 kg

### 1.4 A car survey – a column graph

In this activity sheet, students are given a horizontal column graph. They are asked to read and interpret data from this graph.

**Answers**

**1** **a** 8 **b** 10 **c** 6 **d** 13 **e** 40 **f** 21

**2** White **3** Black **4** 98

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# Teacher's notes

**CONTINUED**

### 1.5 Year 8 classes – a column graph

This activity sheet shows a double (or side-by-side) column graph. Students are asked to read and interpret data from this graph.

**Answers**

**1** **a** 14 **b** 12 **c** 13 **2** **a** 14 **b** 15 **c** 13

**3** 8 M **4** 8 A **5** 8 T **6** 8M and 8T

**7** 8A and 8T **8** **a** 28 **b** 26 **c** 27 **9** 135

### 1.6 Divided bar graphs

This activity sheet shows two divided bar graphs. Both graphs are 100 mm long so each millimetre represents \$1 or 1 student. Students will need to use a ruler to measure the length of each section.

**NOTE:** Photocopying and printing copies from the CD may change the length of some lines and the size of diagrams. When students are asked to measure lengths, it is recommended that the teacher checks the answers using the same copy that is given to students.

**Answers**

**1** **a** Clothes **b** Sport **c** \$40 **d** \$30 **e** \$20

**2** **a** Surfing **b** Cricket and football **c** 25 **d** 15 **e** $\frac{1}{4}$

### 1.7 Summer sports survey – a sector graph

In this activity sheet, students are given a simple sector graph. A sector angle of 1 degree represents one person in the survey. Students will need a protractor to measure the angles in each sector. They are then asked to determine what fraction of the students surveyed voted for various sports.

**Answers**

**1** Surfing

**2** **a** 120° **b** 45° **c** 50° **d** 60° **e** 25° **f** 40° **g** 20°

**3** **a** 120 **b** 50 **c** 60

**4** **a** $\frac{60}{360} = \frac{1}{6}$ **b** $\frac{120}{360} = \frac{1}{3}$ **c** $\frac{40}{360} = \frac{1}{9}$ **d** $\frac{45}{360} = \frac{1}{8}$

### 1.8 Daily travel – a sector graph

This activity sheet shows a sector graph with the results of a survey of 120 students. On this graph, each 3 degrees of a sector angle represents 1 student in the survey. Students are asked to read and interpret data from this graph.

**Answers**

**1** **a** 90° **b** 60°

**2** **a** $\frac{90}{360} = \frac{1}{4}$ **b** $\frac{60}{360} = \frac{1}{6}$ **c** $\frac{30}{360} = \frac{1}{12}$ **d** $\frac{120}{360} = \frac{1}{3}$

**3** 60° **4** **a** 30 **b** 20 **c** 10 **d** 40 **5** 20

### 1.9 Dice results – a line graph

In this activity sheet, students are given a line graph (a frequency polygon) showing the results of an experiment in which a pair of dice was thrown 84 times. They are asked to read and interpret data from this graph.

**Answers**

**1** 7 **2** 14 **3** 9 **4** 2 **5** 41 **6** 43 **7** 28 **8** 11 **9** 84

### 1.10 Temperature graph – a line graph

In this activity sheet, students are given a line graph showing the temperature over a 24 hour period. They are asked to read and interpret data from this graph.

**Answers**

**1** **a** 16° **b** 10° **c** 19° **2** **a** 8° **b** 6 am

**3** **a** 24° **b** 2 pm **4** 6 am and 2 pm

**5** **a** 8 am to 10 am **b** 8 pm to 10 pm

### 1.11 A travel graph – a line graph

This activity sheet shows a travel graph (a type of line graph). Draw attention to the label on the vertical axis. The graph shows *the distance from home*, **not** *the total distance travelled*. Travel graphs may show either variable. Students must read the axes carefully. They are asked to read and interpret data from this graph.

**Answers**

**1** **a** 10 am **b** 12 noon **c** 2 pm **d** 5 pm

**2** 2 hours **3** 1 hour **4** 7 hours **5** 90 km

**6** 60 km **7** 180 km

## EXTENSION IDEAS

Have students interpret graphs found in various media, including the Internet. However, note that graphs sourced from the Internet may not be constructed accurately. The websites of organisations such as the Australian Bureau of Statistics, the Australian Bureau of Meteorology and the United Nations contain many graphs.

Compare different graphs constructed from the same data. Discuss the advantages and disadvantages of different representations of the same data. Ask students which type of graph they think is the best choice for displaying that data.

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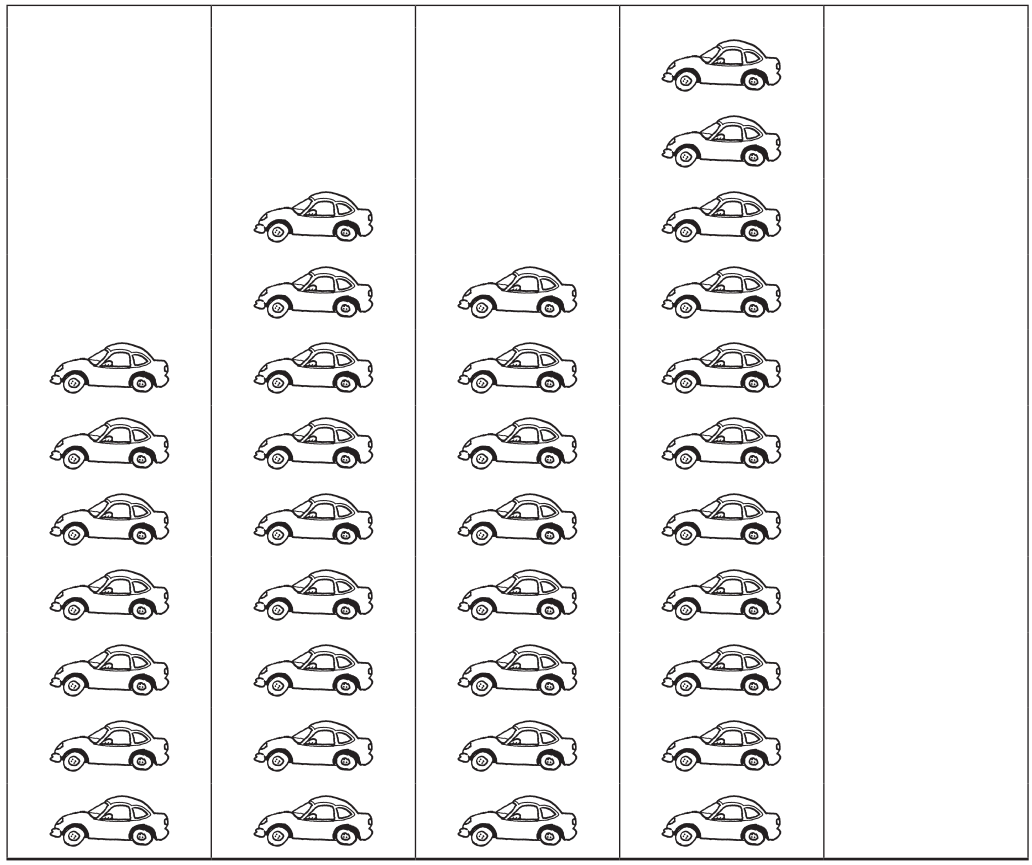
# 1.1 Car sales – a picture graph

This graph shows the number of cars sold over four weeks at a car sales yard.
A drawing of a car is used to represent each car sold.

## Number of cars sold




**Key:** represents 1 car

**1** How many cars were sold during:

**a** the 1st week? _____ **b** the 2nd week? _____

**2** In which week was the **smallest** number of cars sold? _____

**3** In which week was the **largest** number of cars sold? _____

**4** What was the **total** number of cars sold over the four weeks? _____

**5** How many **more** cars were sold in the 4th week than in the 3rd week? _____

**6** In the 5th week, 8 cars were sold. Draw cars in the last column of this graph to show the number of cars sold in the 5th week.

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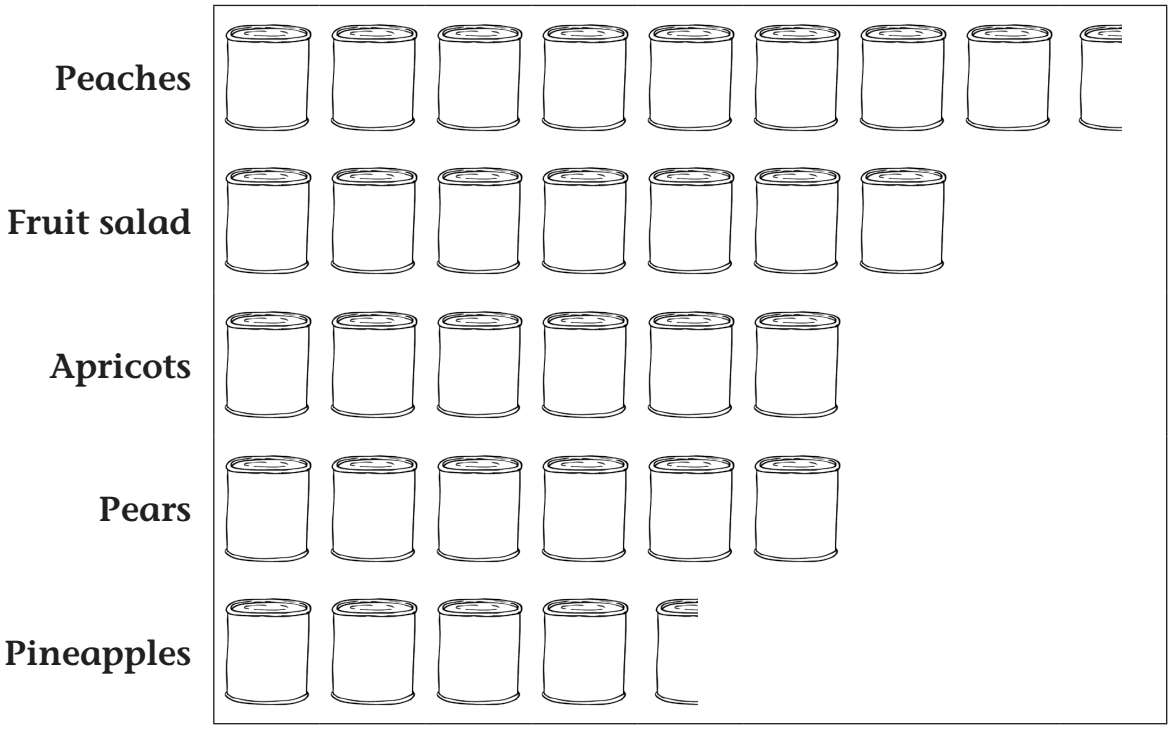
# 1.2 Canned fruit – a picture graph

Peter works part-time at a local supermarket.
He prepared this picture graph for a school project.
It shows the number of cans of fruit sold last week.

## Sales of canned fruit




**1** How many cans of fruit does the symbol represent? _____

**2** How many cans of fruit does the symbol represent? _____

**3** How many cans of fruit salad were sold in the week? _____

**4** How many cans of peaches were sold in the week? _____

**5** Which type of canned fruit had the least sales? _____

**6** Which type of canned fruit had the most sales? _____

**7** How many **more** cans of apricots than cans of pineapples were sold? _____

**8** How many cans of fruit were sold altogether? _____

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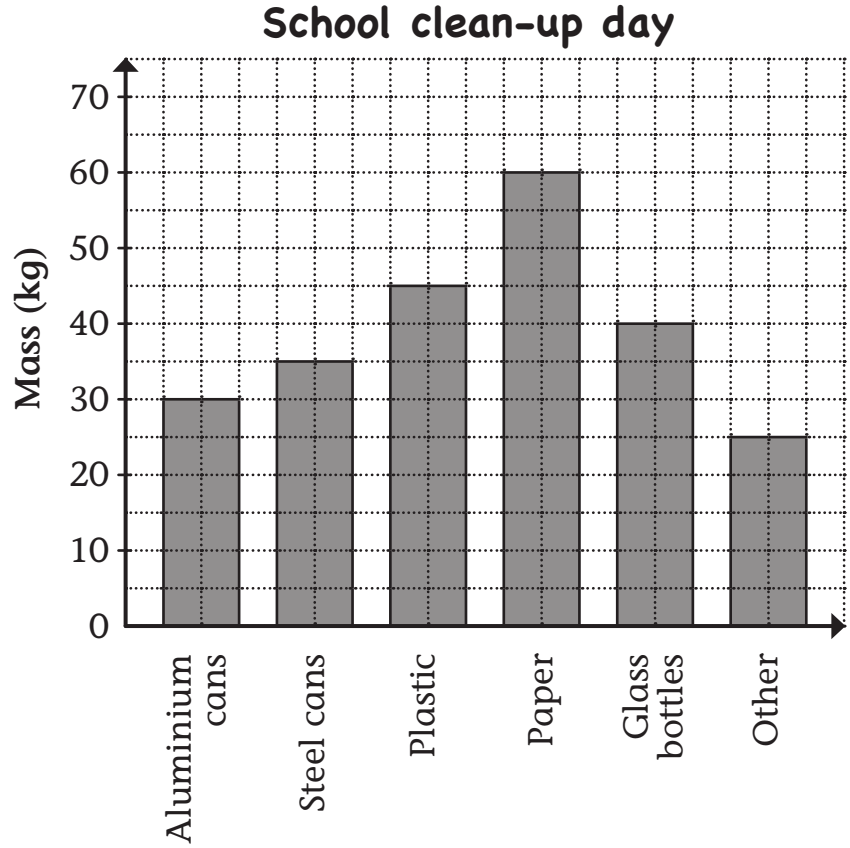
# 1.3 Clean-up day results – a column graph

After a school clean-up day, the items collected were sorted and weighed.

Gillian drew this **column graph** to show the results.

- There is a column for each type of item.
- The height of the column indicates the mass collected.
- Each column is the same width.




**Mass** measures the **amount of matter** in an object.
It is the correct term for what many people call 'weight'.

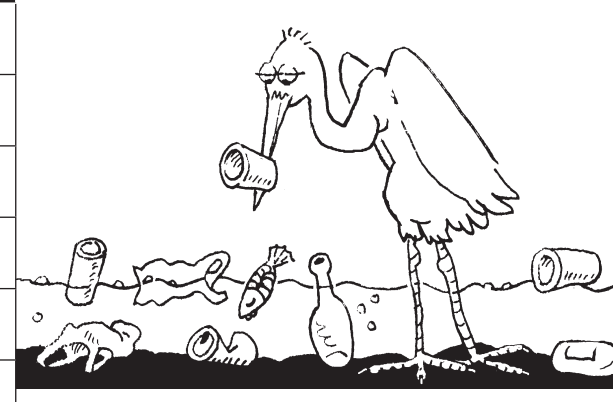
**1** Which type of item had the largest total mass? _____

**2** What mass of aluminium cans was collected? _____

**3** What mass of plastic was collected? _____

**4** Complete this table to show the mass of each type of item collected.

| Type of item | Mass collected |
|---|---|
| Aluminium cans | |
| Steel cans | |
| Plastic | |
| Paper | |
| Glass bottles | |
| Other | |



**5** What was the total mass of the items collected? _____

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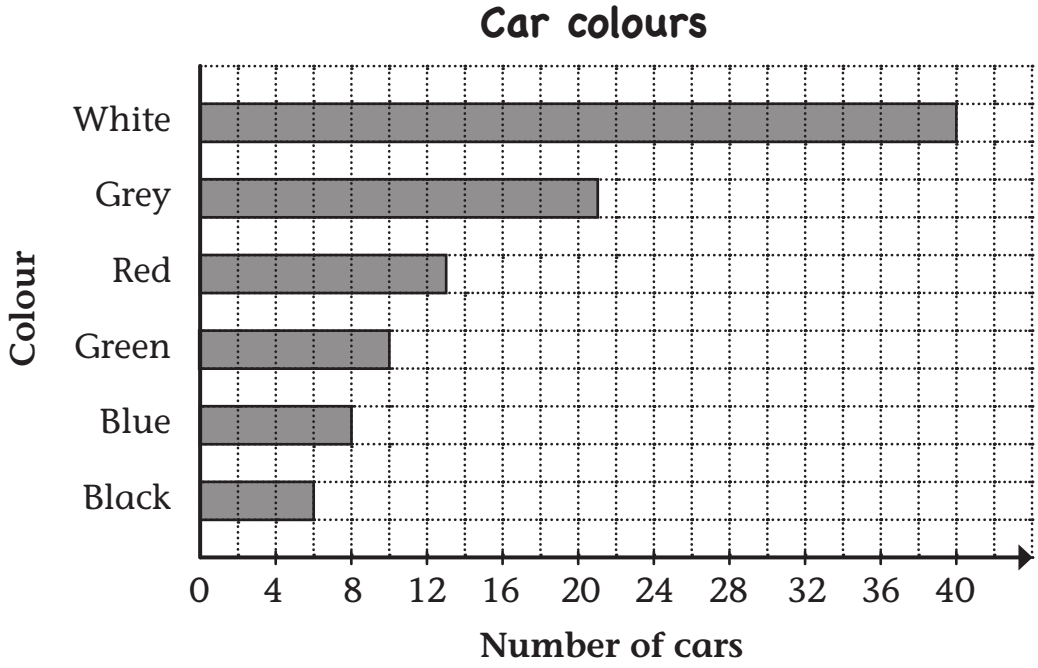
# 1.4 A car survey – a column graph

A class conducted a survey of the cars passing the school.
Marianne recorded the colour of each car.
She presented the results in this **horizontal column graph**.




Tall column graphs can be turned sideways like this graph.
This is often done when some of the columns are very long.

**1** How many cars passing the school were:

**a** blue? _____ **b** green? _____

**c** black? _____ **d** red? _____

**e** white? _____ **f** grey? _____

**2** Which car colour was the **most** common?

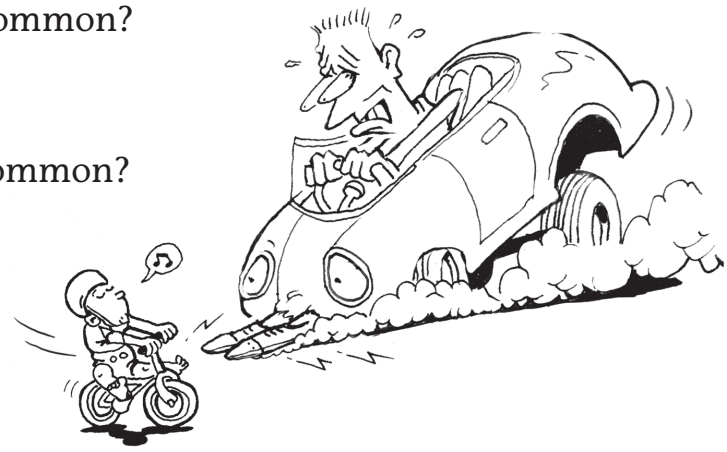
_____

**3** Which car colour was the **least** common?

_____

**4** How many cars altogether did Marianne record in the survey?

_____



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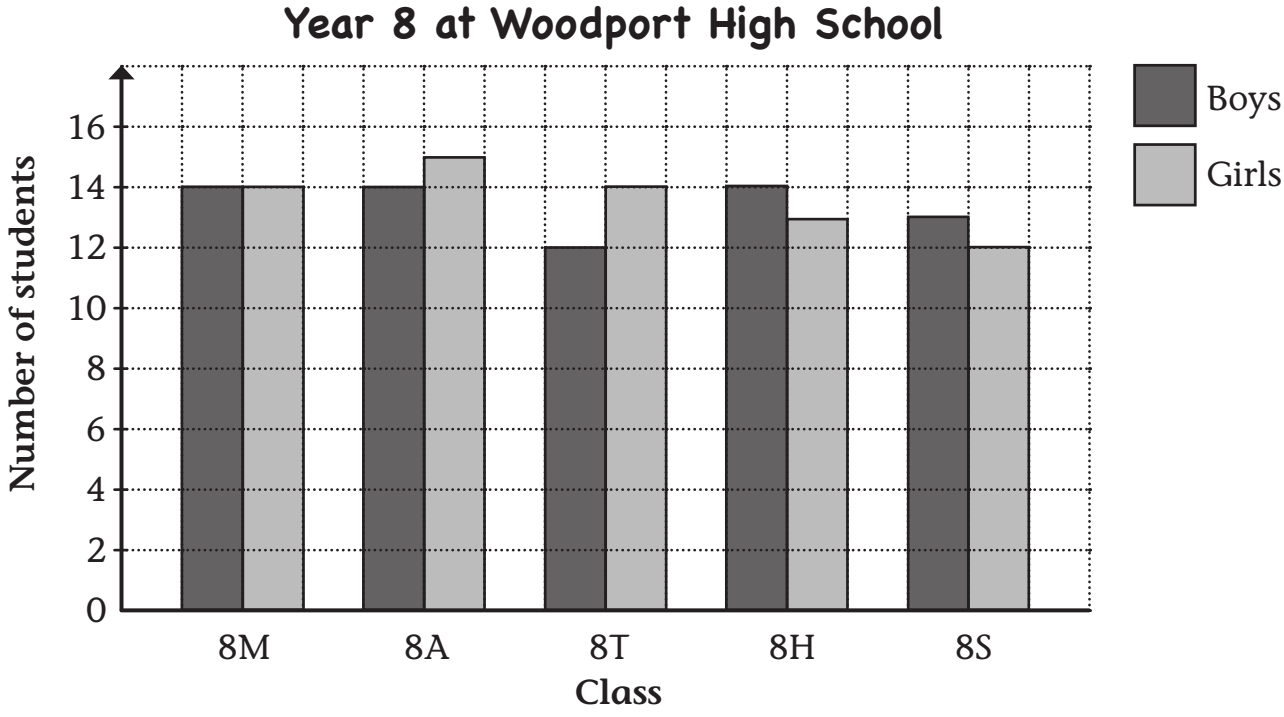
# 1.5 Year 8 classes – a column graph

This **column graph** shows the number of boys and girls in five Year 8 classes at Woodport High School.

The number of boys and girls are shown in separate columns.




**1** How many boys are in:

**a** 8M? _____ **b** 8T? _____ **c** 8S? _____

**2** How many girls are in:

**a** 8M? _____ **b** 8A? _____ **c** 8H? _____

**3** Which class has the **same** number of boys and girls? _____

**4** Which class has the **largest** number of girls? _____

**5** Which class has the **smallest** number of boys? _____

**6** Which two classes have the **same** number of girls? _____

**7** Which two classes have **more** girls than boys? _____

**8** What is the total number of students in:

**a** 8M? _____ **b** 8T? _____ **c** 8H? _____

**9** What is the **total** number of students in Year 8 at Woodport High School? _____

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# 1.6 Divided bar graphs

**1** This graph shows how Nadine spent her pocket money ($100) last month.

## Nadine's pocket money spending

| clothes | school canteen | sport | social activities |
|---|---|---|---|

A **divided bar graph** is a rectangle.
It has been sliced along its length to show how something was divided into parts.
Sometimes, the sections are coloured or shaded differently.

This graph is 100 mm long. Each 1 mm represents $1.

**a** Which is the largest section of the graph? _____

**b** Which is the smallest section of the graph? _____

**c** How much did Nadine spend on clothes? _____

**d** How much did she spend on social activities? _____

**e** How much did she spend at the school canteen? _____

**2** A group of 100 Year 8 boys were asked to nominate their favourite sport.
The results are shown in this divided bar graph.

## Favourite sport of Year 8 boys

| surfing | cricket | football | basketball | other |
|---|---|---|---|---|

**a** Which sport was the most popular? _____

**b** Which two sports received the **same** number of votes? _____

**c** How many boys voted for cricket? _____

**d** How many boys voted for basketball? _____

**e** What **fraction** of the boys voted for football? _____

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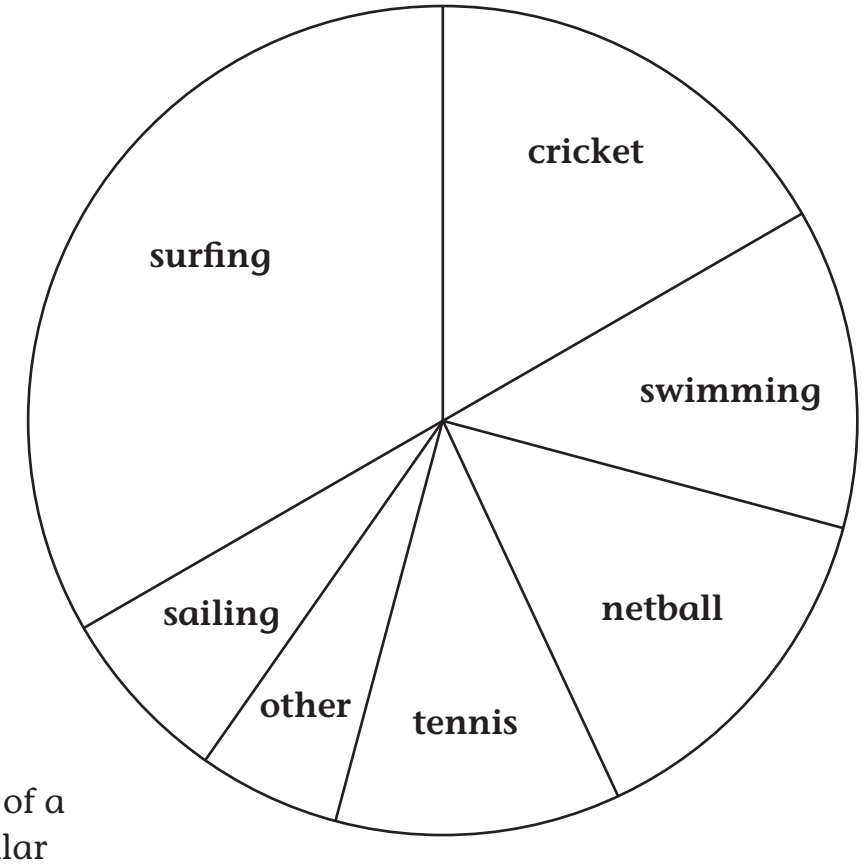
# 1.7 Summer sports survey – a sector graph

Here is a **sector graph**.

The graph looks like a pie or pizza that has been cut into a number of pieces.

Each part is called a **sector**.

Sometimes people call these graphs 'pie charts'.




This graph shows the results of a survey to find the most popular summer sport at Woodport High School.

**1** Name the most popular sport. _____

**2** Use a protractor to **measure** the angle in each sector.

**a** surfing: _____ **b** swimming: _____ **c** netball: _____

**d** cricket: _____ **e** sailing: _____ **f** tennis: _____

**g** other: _____

***Check that your answers add to give 360°.***

**3** If 360 students were surveyed, **how many** voted for:

**a** surfing? _____ **b** netball? _____ **c** cricket? _____

**4** What **fraction** of the students voted for:

**a** cricket? $\frac{60}{360} = \frac{1}{6}$ **b** surfing? $\frac{\quad}{360} =$ _____

**c** tennis? _____ **d** swimming _____

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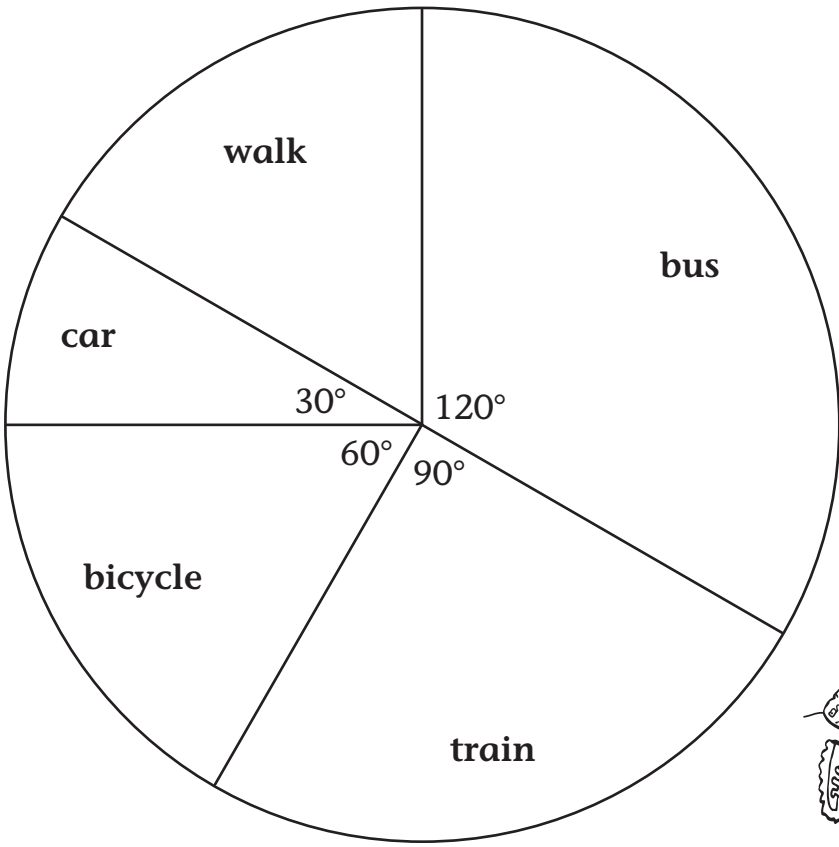
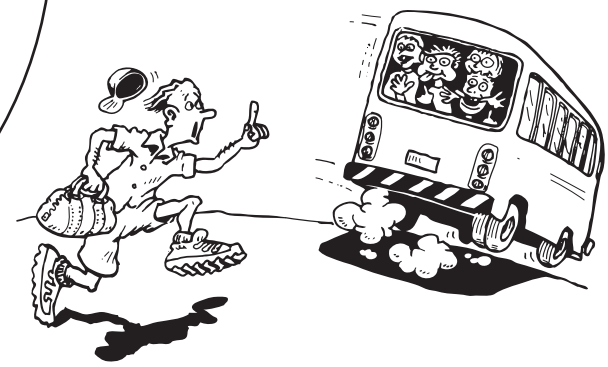
# 1.8 Daily travel – a sector graph

## Method of travel to school




Remo conducted a survey of 120 students to find out how they travel to school each day.

He presented the results in this **sector graph**.



**1** What is the angle in the sector representing the number of students who travel to school by:

**a** train? _____ **b** bicycle? _____

**2** What fraction of the students surveyed travel to school by:

**a** train? $\frac{90}{360} = \frac{1}{4}$ **b** bicycle? _____

**c** car? _____ **d** bus? _____

**3** What is the angle in the sector representing the number of students who walk to school? _____

**4** How many of the 120 students travel to school by:

**a** train? _____ **b** bicycle? _____

**c** car? _____ **d** bus? _____

**5** How many of the 120 students walk to school? _____

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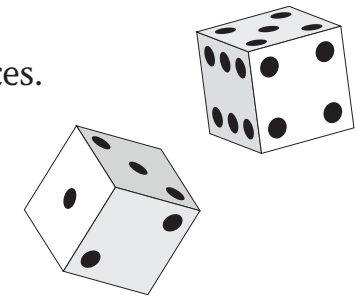
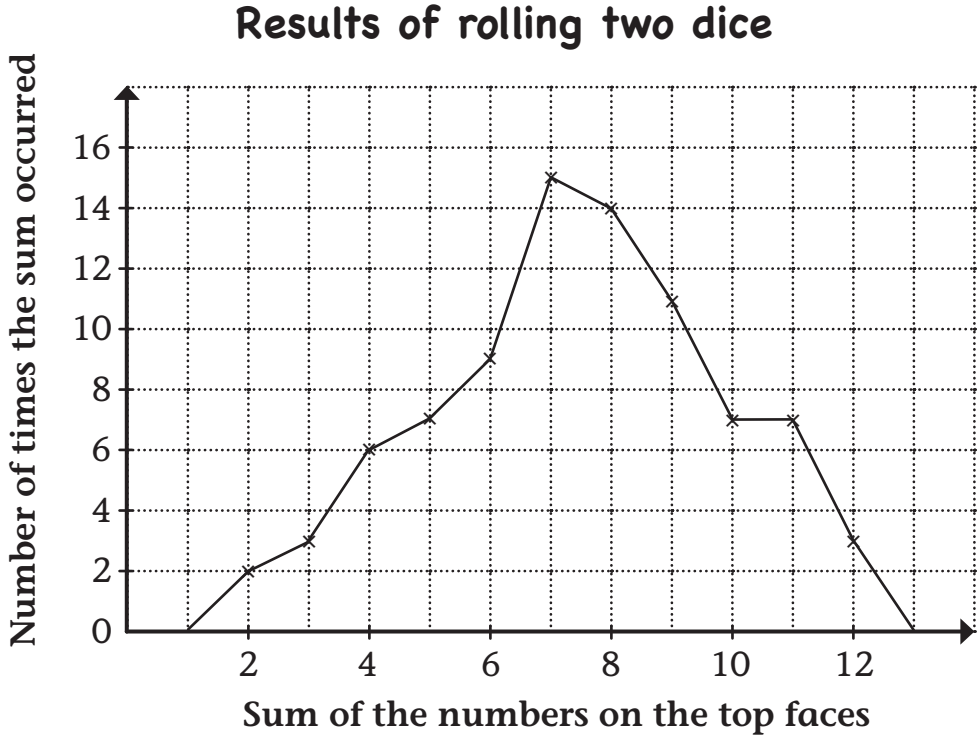
# 1.9 Dice results – a line graph

Beth rolled two dice.
Binh recorded the sum of the two numbers on the top faces.
They repeated this experiment a number of times.
The results are shown in this **line graph**.






**1** Which sum occurred **most** often? _____

**2** How many times was a **sum of 8** obtained? _____

**3** How many times was a **sum of 6** obtained? _____

**4** Which sum occurred **least** often? _____

**5** How many times did an **even sum** occur? _____

**6** How many times did an **odd sum** occur? _____

**7** How many times did a **sum of 9 or more** occur? _____

**8** How many times was the sum **less than 5**? _____

**9** How many times did Beth roll the two dice? _____

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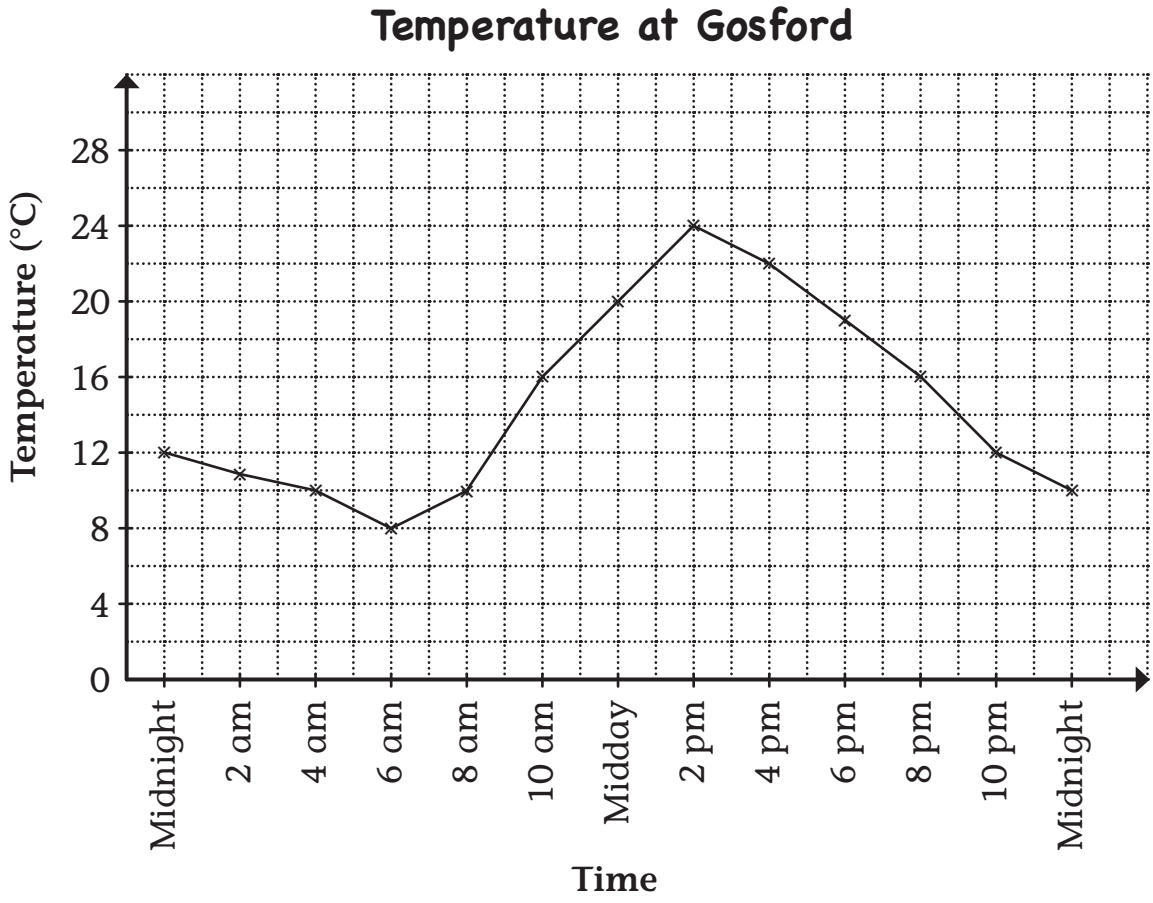
# 1.10 Temperature graph – a line graph

The temperature was recorded at Gosford over a period of 24 hours.
A measurement was recorded every two hours.
The results are shown in this **line graph**.




**1** What temperature was recorded at:

**a** 10 am? _____ **b** 4 am? _____ **c** 6 pm? _____

**2** **a** What was the **minimum** temperature recorded? _____

**b** At what time was this temperature recorded? _____

**3** **a** What was the **maximum** temperature recorded? _____

**b** At what time was this temperature recorded? _____

**4** Between which two times was the temperature **rising**? _____

**5** During which two hour period:

**a** did the temperature **rise** most rapidly? _____

**b** did the temperature **drop** most rapidly? _____

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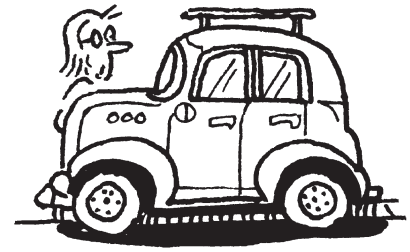
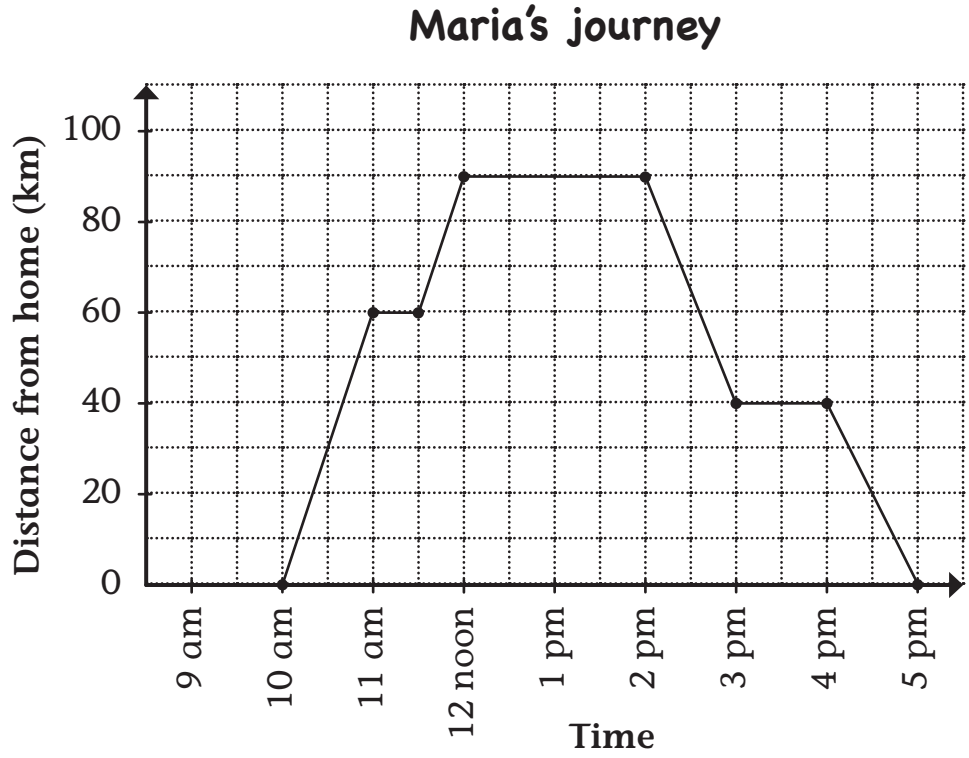
# 1.11 Travel graph – a line graph

Maria went to visit her friend Josie who lived in Wombat.
She stopped to buy petrol on the way.
Maria had lunch with Josie before setting off for home.
She stopped on the way home to do some shopping.

This **line graph** shows how far Maria is from her home.
The horizontal sections show when she stopped.






**1** At what time did Maria:

**a** start her journey? _____ **b** reach Wombat? _____

**c** leave to travel home? _____ **d** return home? _____

**2** For how long did Maria stay in Wombat? _____

**3** For how long did she stop on the way home? _____

**4** For how long was Maria away from home? _____

**5** What is the distance from Maria's home to Wombat? _____

**6** How far from home was Maria when she stopped to buy petrol? _____

**7** What is the total distance Maria travelled? _____

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# Teacher's notes

## OBJECTIVES

- Identify special groups of numbers including odd numbers, even numbers, prime numbers, square numbers and triangular numbers.
- Recognise properties of special groups of whole numbers.
- Calculate and recognise multiples of whole numbers.
- Apply tests for divisibility.
- Recognise the link between square numbers and their square roots.

## PRIOR KNOWLEDGE

Students should be able to perform the basic operations with whole numbers, fractions and decimals using mental arithmetic or a calculator.

## BACKGROUND

In mathematics, words such as *odd*, *product*, and *prime* have different meanings from their everyday usage. You may need to explain these differences for some students.

One is **not** a prime number because it has only one factor, itself.

The square root sign signifies a positive number (or zero). Thus $\sqrt{4} = 2$ only. However, there are two numbers that have a square of 4, namely 2 and -2.

## STARTER ACTIVITY

Ask students to count forwards by twos, starting with 2. Ask what they recognise about these numbers. (They are divisible by 2 or they are even.) Now ask them to count forwards by twos, starting with 1. Ask what they recognise about these numbers. (They are not divisible by 2 or they are odd.)

## ACTIVITY SHEETS

### 2.1 Odd and even numbers

In this activity sheet, students identify even and odd numbers, either using division by 2, or checking the last digit. On the second page they look at some of the properties of even and odd numbers.

**Answers**

**1** 10, 12, 14, 16, 18, 20 **2** 9, 11, 13, 15, 17, 19
**3** 26, 36, 44, 38, 30, 42 **4** 17, 35, 21, 29, 37, 43
**5** **a** 2, 4, 6, 8 or 0 **b** 1, 3, 5, 7 or 9
**6** **a** odd **b** even **c** even **d** odd **e** even **f** odd **g** even **h** even **i** odd **j** odd **k** even **l** even
**7** **a** 24 **b** 17 **c** 31 **d** 40
**8** **a** 16, 18, 20 **b** 21, 23, 25
**9** **a** 14 **b** 18 **c** 20 **d** 24 **e** 10 **f** 20 **g** 30 **h** 34
**10** **a** even **b** even
**11** **a** 11 **b** 15 **c** 21 **d** 27 **e** 13 **f** 15 **g** 23 **h** 35
**12** **a** odd **b** odd
**13** **a** 24 **b** 16 **c** 60 **d** 96 **e** 21 **f** 45 **g** 77 **h** 135
**14** **a** even **b** odd
**15** **a** 18 **b** 40 **c** 90 **d** 84 **e** 20 **f** 42 **g** 88 **h** 90
**16** **a** even **b** even

### 2.2 Multiples

This activity sheet first looks at the concept of a multiple of a whole number. Students are asked to calculate and list the first 12 multiples of the numbers 1 to 10.

**Answers**

**3** 3, 6, 9, 12
**4** 2, 4 ,6, 8, 10, 12, 14, 16, 18, 20
**5** 5, 10, 15, 20, 25, 30, 35, 40, 45, 50
**6** 24, 28, 32
**7**

| | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Multiples of 3 | 3 | 6 | 9 | 12 | 15 | 18 | 21 | 24 | 27 | 30 | 33 | 36 |
| Multiples of 4 | 4 | 8 | 12 | 16 | 20 | 24 | 28 | 32 | 36 | 40 | 44 | 48 |
| Multiples of 5 | 5 | 10 | 15 | 20 | 25 | 30 | 35 | 40 | 45 | 50 | 55 | 60 |
| Multiples of 6 | 6 | 12 | 18 | 24 | 30 | 36 | 42 | 48 | 54 | 60 | 66 | 72 |
| Multiples of 7 | 7 | 14 | 21 | 28 | 35 | 42 | 49 | 56 | 63 | 70 | 77 | 84 |
| Multiples of 8 | 8 | 16 | 24 | 32 | 40 | 48 | 56 | 64 | 72 | 80 | 88 | 96 |
| Multiples of 9 | 9 | 18 | 27 | 36 | 45 | 54 | 63 | 72 | 81 | 90 | 99 | 108 |
| Multiples of 10 | 10 | 20 | 30 | 40 | 50 | 60 | 70 | 80 | 90 | 100 | 110 | 120 |

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# Teacher's notes

CONTINUED

**8** 6, 12, 18 **9** 7, 14, 21, 28

**10** 9, 18, 27, 36 **11** 36, 42, 48

**12** 45, 50, 55, 60, 65, 70

**13** 32, 48, 56 **14** 12, 14, 16, 18

**15** 60, 70, 80, 90 **16** 9 **17** 1

## 2.3 Prime numbers

In this activity sheet, students identify prime numbers less than 120 by a process of eliminating multiples (often referred to as the Sieve of Eratosthenes).

**Answers**

**13** 13

**15** 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97, 101, 103, 107, 109, 113

## 2.4 Divisibility and factors

In this activity sheet, students are first reminded of the divisibility tests for numbers divisible by 2, 3, 4, 5, 6, 8, 9 and 10. (There is no test for divisibility by 7.) They then use these tests to check for factors of whole numbers.

**Answers**

**1** 75, 249, 438, 516, 8 229

**2** 116, 348, 552, 1 496, 3 116

**3** 65, 90, 370, 765, 1 395, 2 630

**4** 84, 96, 636, 1 266, 5 610, 7 398

**5** 198, 252, 747, 2 106, 3 519, 9 378

**6** **a** Yes **b** No **c** No **d** Yes **e** Yes **f** Yes **g** Yes **h** No **i** No **j** Yes

**7** **a** True **b** False **c** True **d** False **e** False **f** True **g** True **h** True **i** True **j** False

**8** **a** No **b** Yes **c** No **d** Yes **e** No **f** Yes **g** Yes **h** No **i** No **j** Yes

## 2.5 Square numbers

Dot diagrams are used to introduce the pattern of square numbers. Students are then asked to calculate the squares of various numbers. Calculators will be useful.

**Answers**

**2** $7^2 = 49$ **3** 1, 4, 9, 16, 25, 36, 49, 64, 81, 100

**4** $15^2 = 225$

**5** **a** 64 **b** 100 **c** 81 **d** 144 **e** 196 **f** 256 **g** 324 **h** 400 **i** 625 **j** 484 **k** 576 **l** 729 **m** 900 **n** 2 500 **o** 10 000

## 2.6 Square roots

This activity sheet develops the link between square numbers and their square roots. Students are then asked to calculate the square root of various numbers. In the last question they will need to round their answers to 1 decimal place. Calculators will be needed.

**Answers**

**1** **b** $3^2 = 3 \times 3 = 9$ so $\sqrt{9} = 3$
**c** $4^2 = 4 \times 4 = 16$ so $\sqrt{16} = 4$
**d** $5^2 = 5 \times 5 = 25$ so $\sqrt{25} = 5$
**e** $6^2 = 6 \times 6 = 36$ so $\sqrt{36} = 6$
**f** $9^2 = 9 \times 9 = 81$ so $\sqrt{81} = 9$
**g** $12^2 = 12 \times 12 = 144$ so $\sqrt{144} = 12$

**2** **a** 5 **b** 6 **c** 7 **d** 8 **e** 10 **f** 13 **g** 15 **h** 16 **i** 20

**3** **a** 3.9 **b** 4.5 **c** 5.1 **d** 5.5 **e** 6.5 **f** 8.4 **g** 11.2 **h** 12.2 **i** 14.1

## 2.7 Triangular numbers

The arrangements of ten bowling pins and 15 pool balls are used to introduce triangular numbers. Students use dot diagrams and a pattern of number sums to calculate the first ten triangular numbers.

**Answers**

**1** 15

**3** $1 + 2 + 3 + 4 + 5 = 15$
$1 + 2 + 3 + 4 + 5 + 6 = 21$
$1 + 2 + 3 + 4 + 5 + 6 + 7 = 28$

**4** 1, 3, 6, 10, 15, 21, 28, 36, 45, 55

# EXTENSION IDEAS

Investigate other special groups of numbers, such as palindromic numbers (numbers that remain the same when the order of the digits is reversed; e.g. 141, 8 448, 12 621) and Fibonacci numbers (a number sequence formed by adding the previous two terms; 1, 1, 2, 3, 5, 8, 13 ...)

Encourage students to apply the tests of divisibility when determining factors and as an aid to calculation.

Students could find the Highest Common Factor of two given numbers after listing their factors.

Students could calculate cubes and related cube roots of numbers.

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# 2.1 Odd and even numbers

A **remainder** is the amount left over in a division.

An **even number** has no remainder when it is divided by 2.
We say an even number is divisible by 2.

An **odd number** has a remainder of 1 when it is divided by 2.
An odd number is ***not*** divisible by 2.

**1** The first four **even numbers** are **2**, **4**, **6** and **8**. Write the *next* six even numbers.

_____ _____ _____ _____ _____ _____

**2** The first four **odd numbers** are **1**, **3**, **5** and **7**. Write the *next* six odd numbers.

_____ _____ _____ _____ _____ _____

**3** Circle the **even numbers**.

26 36 21 44 19 38 30 43 37 42

**4** Circle the **odd numbers**.

17 22 35 40 21 26 29 37 48 43

**5** Look at your answers to questions 1 to 4 and complete these statements.

**a** An **even number** must have a final digit which is ___, ___, ___, ___ or ___.

**b** An **odd number** must have a final digit which is ___, ___, ___, ___ or ___.

**6** Are the following numbers **even** or **odd**?

**a** 15 _____ **b** 24 _____ **c** 86 _____

**d** 93 _____ **e** 106 _____ **f** 159 _____

**g** 332 _____ **h** 256 _____ **i** 953 _____

**j** 1 235 _____ **k** 4 636 _____ **l** 7 398 _____

**7** Write:

**a** the next even number after 22. ____ **b** the next odd number after 15. ____

**c** the next odd number after 29. ____ **d** the next even number after 38. ____

**8** Write:

**a** the next three **even** numbers after **14**. _____ _____ _____

**b** the next three **odd** numbers after **19**. _____ _____ _____

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# 2.1 Odd and even numbers

**CONTINUED**

**9** Complete these additions.

**a** 6 + 8 = _____ **b** 8 + 10 = _____ **c** 14 + 6 = _____ **d** 16 + 8 = _____

**e** 7 + 3 = _____ **f** 11 + 9 = _____ **g** 13 + 17 = _____ **h** 19 + 15 = _____

**10** Use the words *even* or *odd* to complete these statements.

**a** An even number plus an even number makes an _____ number.

**b** An odd number plus an odd number makes an _____ number.

**11** Complete these additions.

**a** 6 + 5 = _____ **b** 8 + 7 = _____ **c** 12 + 9 = _____ **d** 16 + 11 = _____

**e** 5 + 8 = _____ **f** 9 + 6 = _____ **g** 15 + 8 = _____ **h** 19 + 16 = _____

**12** Use the words *even* or *odd* to complete these statements.

**a** An even number plus an odd number makes an _____ number.

**b** An odd number plus an even number makes an _____ number.

**13** Complete these multiplications.

**a** 4 × 6 = _____ **b** 8 × 2 = _____ **c** 6 × 10 = _____ **d** 12 × 8 = _____

**e** 7 × 3 = _____ **f** 9 × 5 = _____ **g** 11 × 7 = _____ **h** 15 × 9 = _____

**14** Use the words *even* or *odd* to complete these statements.

**a** An even number times an even number makes an _____ number.

**b** An odd number times an odd number makes an _____ number.

**15** Complete these multiplications.

**a** 6 × 3 = _____ **b** 8 × 5 = _____ **c** 10 × 9 = _____ **d** 12 × 7 = _____

**e** 5 × 4 = _____ **f** 7 × 6 = _____ **g** 11 × 8 = _____ **h** 15 × 6 = _____

**16** Use the words *even* or *odd* to complete these statements.

**a** An even number times an odd number makes an _____ number.

**b** An odd number times an even number makes an _____ number.

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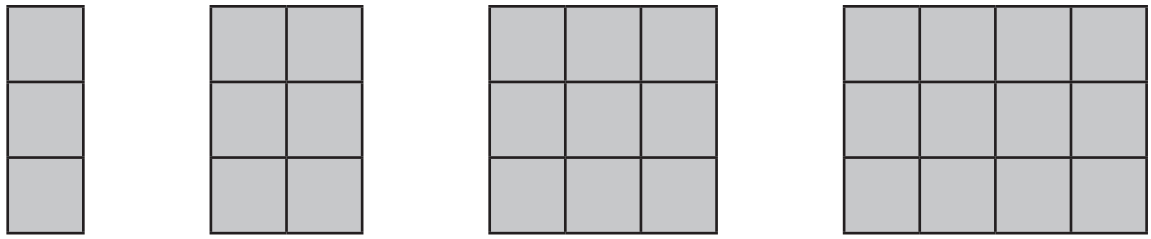
# 2.2 Multiples

**1** Place centicubes or multi-link cubes in blocks of 3, or triples, as shown at the right.

**2** Place one triple on its own. Place two triples together. Now place three triples together and form this pattern:



____ ____ ____ ____

**3** Count the number of cubes in each shape. Write your answers on the lines below each shape.

You have built models of the first four **multiples of 3**.

$3 \times 1 = 3$ $3 \times 2 = 6$ $3 \times 3 = 9$ $3 \times 4 = 12$

A **multiple** of a number is the product of that number and any other whole number.

The first ten multiples of 3 are:

| $3 \times 1$ | $3 \times 2$ | $3 \times 3$ | $3 \times 4$ | $3 \times 5$ | $3 \times 6$ | $3 \times 7$ | $3 \times 8$ | $3 \times 9$ | $3 \times 10$ |
|---|---|---|---|---|---|---|---|---|---|
| 3 | 6 | 9 | 12 | 15 | 18 | 21 | 24 | 27 | 30 |

Many students memorise the first ten multiples of the numbers from 1 to 10 when they are 'learning their tables'.

**4** Write the first ten multiples of 2.

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | |

**5** Write the first ten multiples of 5.

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | |

**6** Write the next three multiples of 4, after 20.

4, 8, 12, 16, 20, _____, _____, _____

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# 2.2 Multiples

**CONTINUED**

**7** Complete this table of the first twelve multiples of the numbers from 2 to 10.

| | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Multiples of 2** | 2 | 4 | 6 | 8 | 10 | 12 | 14 | 16 | 18 | 20 | 22 | 24 |
| **Multiples of 3** | 3 | 6 | 9 | 12 | 15 | 18 | 21 | | | | | |
| **Multiples of 4** | 4 | 8 | 12 | 16 | 20 | 24 | | | | | | |
| **Multiples of 5** | 5 | 10 | 15 | 20 | | | | | | | | |
| **Multiples of 6** | | | | | | | | | | | | |
| **Multiples of 7** | | | | | | | | | | | | |
| **Multiples of 8** | | | | | | | | | | | | |
| **Multiples of 9** | | | | | | | | | | | | |
| **Multiples of 10** | | | | | | | | | | | | |

**Use the table to help you answer the following questions.**

**8** List the multiples of 6 which are less than 20. _____

**9** List the multiples of 7 which are less than 30. _____

**10** List the multiples of 9 which are less than 40. _____

**11** Write the next three multiples of 6 after 30. _____ _____ _____

**12** Write the next six multiples of 5 after 40. _____ _____ _____ _____ _____ _____

**13** Fill in the missing numbers in this pattern of multiples of 8.

8, 16, 24, _____, 40, _____, _____, 64

**14** List the multiples of 2 between 11 and 19. _____

**15** List the multiples of 10 between 51 and 99. _____

**16** How many multiples of 5 are smaller than 50? _____

**17** How many multiples of 9 are between 30 and 40? _____

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# 2.3 Prime numbers

A **prime number** has only two factors – itself *and* 1.
A number with more than two factors is called a c**omposite number**.

To find all the prime numbers less than 120, use the number grid on the next page and follow these steps carefully.

**1** Cross out **1**. It is **not** a prime number. It does not have two factors.

**2** Put a circle around **2**. It is a prime number.

**3** Cross out all the other even numbers. (The last digit will be 2, 4, 6, 8 or 0.)
These are the **multiples of 2**. They are **not** prime numbers.

**4** Put a circle around **3**. It is the next prime number.

**5** Cross out all the **multiples of 3**.
They have more than two factors. They are **not** prime numbers.

**6** Put a circle around **5**. It is the next number not crossed out.
It is the next prime number.

**7** Cross out all the other **multiples of 5**. (The last digit will be 5 or 0.)

**8** Put a circle around the next number not crossed out. It is **7**.
7 is the next prime number.

**9** Cross out all the **multiples of 7**.

**10** Put a circle around the next number not crossed out. It is **11**.
11 is the next prime number.

**12** Cross out all the **multiples of 11**.

**13** What is the next number not crossed out? _____

This is the next prime number. Cross out any of its multiples that remain.

**14** Check that the remaining numbers are prime and circle them.

**15** List all the prime numbers less than 120. (All the numbers you have circled.)
You should have 30 numbers.

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | |
| | | | | | | | | | |
| | | | | | | | | | |

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# 2.3 Prime numbers

CONTINUED

## Number grid

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 |
| 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 |
| 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 |
| 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 |
| 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 |
| 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 |
| 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 |
| 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| 101 | 102 | 103 | 104 | 105 | 106 | 107 | 108 | 109 | 110 |
| 111 | 112 | 113 | 114 | 115 | 116 | 117 | 118 | 119 | 120 |

A mathematician named Goldbach proposed that:

*Every even number, except 2, can be written as the sum of two prime numbers* (which can be the same).

**16** Complete this list by writing each number as the sum of two prime numbers.

4 = 2 + 2

6 = 3 + 3

8 = 3 + 5

10 = 5 + 5

12 = 5 + 7

14 = 7 + 7

16 = _____

18 = _____

20 = _____

22 = _____

24 = _____

26 = _____

28 = _____

30 = _____

32 = _____

34 = _____

36 = _____

38 = _____

40 = _____

42 = _____

44 = _____

46 = _____

48 = _____

50 = _____

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# 2.4 Divisibility and factors

A **factor** is a number that **exactly divides** into another number.
One way to find the factors of a number without doing the actual division is to use the tests for divisibility.

Some simple tests for divisibility are listed below.

A number is:

- **divisible by 2** if the last digit is 0, 2, 4, 6 or 8 (i.e. an even number)
- **divisible by 3** if the sum of its digits is divisible by 3
- **divisible by 4** if the last two digits form a number that is divisible by 4
- **divisible by 5** if the last digit is 5 or 0
- **divisible by 6** if it is divisible by both 2 and 3 (i.e. it is an even number *and* the sum of the digits is divisible by 3)
- **divisible by 8** if the last three digits form a number that is divisible by 8
- **divisible by 9** if the sum of the digits is divisible by 9
- **divisible by 10** if the last digit is 0.

**** There is no simple test for divisibility by 7.***

**1** Apply the *sum of the digits* test and circle the numbers which are **divisible by 3**.

75 89 137 249 376 438 516 754 1 625 8 229

**2** Apply the *last two digits* test and circle the numbers which are **divisible by 4**.

116 234 348 552 747 1 082 1 496 2 002 3 116 4 614

**3** Circle the numbers that are **divisible by 5**.

53 65 90 256 370 765 1 523 1 395 2 630 5 502

**4** Circle the numbers that are **divisible by 6**.

84 96 124 369 636 1 266 1 323 2 351 5 610 7 398

**5** Circle the numbers that are **divisible by 9**.

129 198 252 619 747 1 381 2 106 3 519 7 435 9 378

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# 2.4 Divisibility and factors

**CONTINUED**

**6** Use the divisibility tests to answer the following questions.

**a** Is 578 divisible by 2? _____

**b** Is 1 082 divisible by 4? _____

**c** Is 763 divisible by 3? _____

**d** Is 1 615 divisible by 5? _____

**e** Is 1 848 divisible by 8? _____

**f** Is 2 754 divisible by 9? _____

**g** Is 7 368 divisible by 3? _____

**h** Is 39 047 divisible by 4? _____

**i** Is 60 305 divisible by 10? _____

**j** Is 83 070 divisible by 6? _____

**7** Label each statement as **true** or **false**. Remember, a ***factor*** is a number that exactly divides into another number.

**a** 4 is a factor of 64 _____

**b** 3 is a factor of 52 _____

**c** 5 is a factor of 980 _____

**d** 2 is a factor of 125 _____

**e** 9 is a factor of 218 _____

**f** 6 is a factor of 288 _____

**g** 3 is a factor of 1 080 _____

**h** 5 is a factor of 1 375 _____

**i** 10 is a factor of 5 700 _____

**j** 9 is a factor of 3 545 _____

**8** Answer **yes** or **no**.

**a** Is 4 a factor of 402? _____

**b** Is 2 a factor of 476? _____

**c** Is 3 a factor of 796? _____

**d** Is 5 a factor of 1 870? _____

**e** Is 6 a factor of 842? _____

**f** Is 4 a factor of 1 764? _____

**g** Is 9 a factor of 2 169? _____

**h** Is 6 a factor of 1 106? _____

**i** Is 3 a factor of 5 296? _____

**j** Is 9 a factor of 2 835? _____

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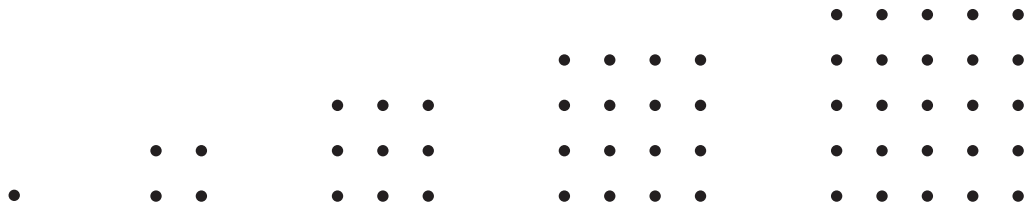
# 2.5 Square numbers

Square numbers can be represented by a pattern of dots.
They are arranged with the same number of dots in each row and each column.



The diagrams above show the first five square numbers.

$1 = 1 \times 1 = 1^2$ (Called **one squared**)
$4 = 2 \times 2 = 2^2$ (Called **two squared**)
$9 = 3 \times 3 = 3^2$ (Called **three squared**)
$16 = 4 \times 4 = 4^2$ (Called **four squared**)
$25 = 5 \times 5 = 5^2$ (Called **five squared**)

The next number in this pattern would be $6^2 = 6 \times 6 = 36$.

**1** Draw a dot diagram in the pattern above to represent 36.

**2** What is the next square number? _____

> ***Rule:*** **To square a number is to multiply it by itself.**
> For example: $8^2 = 8 \times 8 = 64$

**3** Write the values of the first ten square numbers.

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | |

**4** What is the $15^{th}$ square number? _____

**5** Find the value of the following square numbers.

**a** $8^2$ = _____ **b** $10^2$ = _____ **c** $9^2$ = _____

**d** $12^2$ = _____ **e** $14^2$ = _____ **f** $16^2$ = _____

**g** $18^2$ = _____ **h** $20^2$ = _____ **i** $25^2$ = _____

**j** $22^2$ = _____ **k** $24^2$ = _____ **l** $27^2$ = _____

**m** $30^2$ = _____ **n** $50^2$ = _____ **o** $100^2$ = _____

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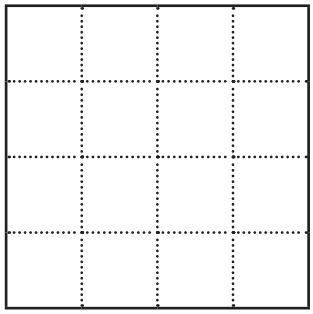
# 2.6 Square roots

Joshua wants to draw a square with an area of 16 cm$^2$.
What is the length of each side of this square?



This square is drawn on 1 cm grid paper.
The length of each side is 4 cm.
$4^2 = 4 \times 4 = 16$
To find the length of each side, find the **square root** of 16.
We write: $\sqrt{16} = 4$
Finding the square root is the **inverse** of squaring a number.

**1** Complete these statements.

**a** $2^2 = 2 \times 2 = 4$ so $\sqrt{4} = 2$

**b** $3^2 = 3 \times 3 = 9$ so $\sqrt{9} =$ _____

**c** $4^2 = 4 \times 4 =$ _____ so $\sqrt{16} =$ ____

**d** $5^2 = 5 \times 5 =$ _____ so $\sqrt{\phantom{00}} = 5$

**e** $6^2 = 6 \times 6 =$ _____ so $\sqrt{\phantom{00}} = 6$

**f** $9^2 = 9 \times 9 =$ _____ so $\sqrt{\phantom{00}} = 9$

**g** $12^2 = 12 \times 12 =$ _____ so $\sqrt{\phantom{00}} = 12$

**2** Find the values of these square roots.

**a** $\sqrt{25} =$ _____ **b** $\sqrt{36} =$ _____

**c** $\sqrt{49} =$ _____ **d** $\sqrt{64} =$ _____

**e** $\sqrt{100} =$ _____ **f** $\sqrt{169} =$ _____

**g** $\sqrt{225} =$ _____ **h** $\sqrt{256} =$ _____

**i** $\sqrt{400} =$ _____

**3** Use your calculator to find the values of these square roots.
Round each answer to 1 decimal place.

**a** $\sqrt{15} =$ _____ **b** $\sqrt{20} =$ _____ **c** $\sqrt{26} =$ _____

**d** $\sqrt{30} =$ _____ **e** $\sqrt{42} =$ _____ **f** $\sqrt{70} =$ _____

**g** $\sqrt{125} =$ _____ **h** $\sqrt{150} =$ _____ **i** $\sqrt{200} =$ _____

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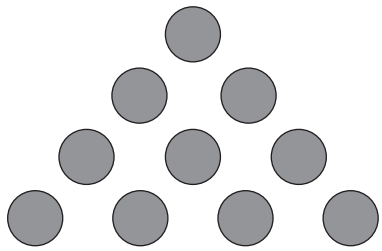
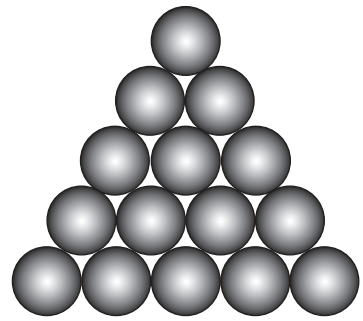
# 2.7 Triangular numbers

Have you ever been ten-pin bowling or played pool?

The 10 pins are arranged in a triangular pattern.



The 15 pool balls start in a triangular pattern.

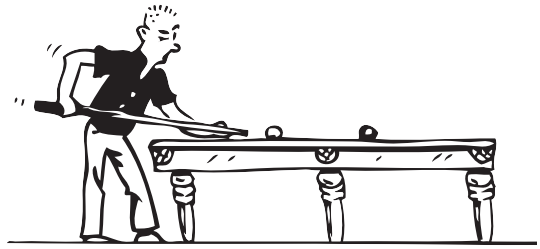


Numbers like 10 and 15 are called **triangular numbers**.
They can be represented by a pattern of dots in the shape of a triangle.
Here are the first four numbers in this pattern.

**1** **3** **6** **10**

**1** What is the next triangular number? _____

**2** Draw the next diagram in the pattern above.



**3** Compare these number sentences with the dot diagrams.
Complete three more lines of this pattern.

$1 = 1$

$1 + 2 = 3$

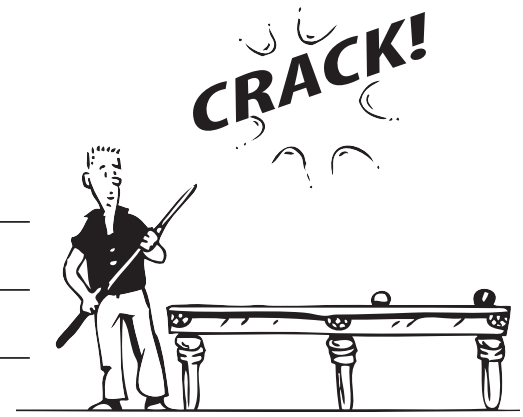
$1 + 2 + 3 = 6$

$1 + 2 + 3 + 4 = 10$

_____

_____

_____




**4** Write the first ten triangular numbers.

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | |

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# Teacher's notes

## OBJECTIVES

- Generate, describe and record geometric patterns using a variety of strategies.
- Generate, describe and record number patterns using a variety of strategies.
- Complete a table of values for geometric and number patterns.
- Extend a number pattern and find a higher term, given the first four terms.
- Describe a number pattern using words.

## PRIOR KNOWLEDGE

Students should be able to perform the basic operations with whole numbers, fractions and decimals using mental arithmetic or a calculator.

## BACKGROUND

These activity sheets may be used to introduce algebra, without using symbols. They provide a variety of geometric patterns so that students see the development of algebra in a variety of situations. Encourage students to use their own words when they describe number patterns. Patterns can usually be described in more than one way. You could discuss students' descriptions and suggest the use of appropriate mathematical language.

## STARTER ACTIVITY

Ask students what comes next in a repeating pattern using familiar objects. For example: blue card, red card, blue card, red card ...; square, triangle, circle, square, triangle ...

## ACTIVITY SHEETS

### 3.1 Building patterns with toothpicks

In this activity sheet, students build patterns using toothpicks, then draw the next step in the pattern. (Matches or paddle-pop sticks could be used in place of toothpicks.) They are asked to count the number of toothpicks used in each step and write their answers in a table.

**Answers**

**2** **a** 2, 3, 4, 5, 6 **b** 2, 4, 6, 8, 10
**c** 4, 7, 10, 13, 16 **d** 4, 6, 8, 10, 12

### 3.2 Building patterns with squares

In this activity sheet, students are given a pattern built by shading squares on a grid. They are asked to draw the next step in the pattern. They are also asked to count the number of shaded squares in each step and write their answers in a table.

**Answers**

**2** **a** 2, 4, 6, 8, 10 **b** 4, 5, 6, 7, 8 **c** 5, 6, 7, 8, 9

**3** **b** 3, 5, 7, 9 **c** 11, 13

**4** **b** 1, 4, 9, 16 **c** 25, 36

### 3.3 Building patterns with cubes

In this activity sheet, students build patterns using cubes, then draw the next step in the pattern. They are asked to count the number of cubes used in each step and write their answers in a table.

**Answers**

**1** 1, 4, 9, 16, 25

**2** 1, 3, 6, 10, 15

**3** 21

**4** 4, 7, 10, 13, 16

**5** **a** 19 **b** 22

**6** 1, 8, 27, 64, 125

**7** 216

### 3.4 Describing geometric patterns

This activity sheet looks at patterns made using toothpicks. Students count the number of toothpicks needed to build each step and write their answers in a table. They are asked to continue the pattern and complete a sentence to describe the pattern. They are also asked to predict the value of a higher term in each pattern.

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# Teacher's notes

**CONTINUED**

**Answers**

**1** **a** 3, 6, 9, 12, 15, 18, 21, 24
**b** three times
**c** **(i)** 60 **(ii)** 150

**2** **a** 4, 7, 10, 13, 16, 19, 22, 25
**b** add one
**c** **(i)** 61 **(ii)** 151

**3** **a** 3, 5, 7, 9, 11, 13, 15, 17
**b** two times
**c** **(i)** 51 **(ii)** 201

**4** **a** 6, 11, 16, 21, 26, 31, 36, 41
**b** five times, add one
**c** **(i)** 151 **(ii)** 226

### 3.5 Using pattern rules

In this activity sheet, students are given a sentence which describes a pattern built using toothpicks. They are asked to find the number of toothpicks needed to build a higher step in the pattern.

**Answers**

**1** **b** 48

**2** **a** 3 **b** 9 **c** 15 **d** 36

**3** **a** 6 **b** 24 **c** 36 **d** 60

**4** **a** 5 **b** 11 c 17 **d** 25

**5** **a** 11 **b** 19 **c** 35 **d** 59

### 3.6 Describing number patterns

In this activity sheet, students are given the first four terms in a selection of number patterns. Each pattern is generated by a single operation (addition, subtraction, multiplication or division). They are asked to continue the pattern and describe the pattern in words. In the last question they are given a starting number and a rule. They are asked to write the next three terms in each pattern.

**Answers**

**1** **b** 13, 16, 19, 22; start with 1, add 3
**c** 25, 30, 35, 40; start with 5, add 5
**d** 22, 26, 30, 34; start with 6, add 4
**e** 22, 27, 32, 37; start with 2, add 5
**f** 40, 49, 58, 67; start with 4, add 9

**2** **b** 6, 5, 4, 3; start with 10, subtract 1
**c** 12, 10, 8, 6; start with 20, subtract 2
**d** 80, 75, 70, 65; start with 100, subtract 5
**e** 20, 16, 12, 8; start with 36, subtract 4
**f** 26, 20, 14, 8; start with 50, subtract 6

**3** **b** 48, 96, 192; start with 3, multiply by 2
**c** 81, 243, 729; start with 1, multiply by 3
**d** 80, 160, 320; start with 5, multiply by 2
**e** 1 250, 6 250, 31 250; start with 2, multiply by 5

**4** **b** 6, 3; start with 96, divide by 2
**c** 3, 1; start with 243, divide by 3
**d** 10, 1; start with 100 000, divide by 10
**e** 15, 5; start with 1 215, divide by 3

**5** **b** 2, 5, 8, 11 **c** 4, 9, 14, 19 **d** 7, 11, 15, 19
**e** 5, 14, 23, 32 **f** 12, 10, 8, 6
**g** 50, 45, 40, 35 **h** 36, 30, 24, 18
**i** 3, 6, 12, 24 **j** 10, 20, 40, 80
**k** 5, 15, 45, 135 **l** 9, 36, 144, 576
**m** 32, 16, 8, 4 **n** 1 000, 200, 40, 8

## EXTENSION IDEAS

Ask students to identify patterns in data displayed in a spreadsheet. These could be as simple as some of the completed tables from activity sheets 3.4 and 4.4.

Ask students to identify a missing term in the middle of a pattern, e.g. 2, 5, 8, __, 14.

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# 3.1 Building patterns with toothpicks

Chloe used toothpicks to build these patterns.

**1** Continue each pattern by drawing the next step.

**a**

**b**

**c**

**d**

**e**

**f**

**g**

**h**

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# 3.1 Building patterns with toothpicks

**CONTINUED**

**2** For each pattern:

- continue it by drawing the next step
- count the number of toothpicks used in each of the first four steps and write your answers in the table provided
- predict the number of toothpicks required to build step 5.

**a**

| Step in the pattern | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| Number of toothpicks | | | | | |

**b**

| Step in the pattern | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| Number of toothpicks | | | | | |

**c**

| Step in the pattern | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| Number of toothpicks | | | | | |

**d**

| Step in the pattern | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| Number of toothpicks | | | | | |

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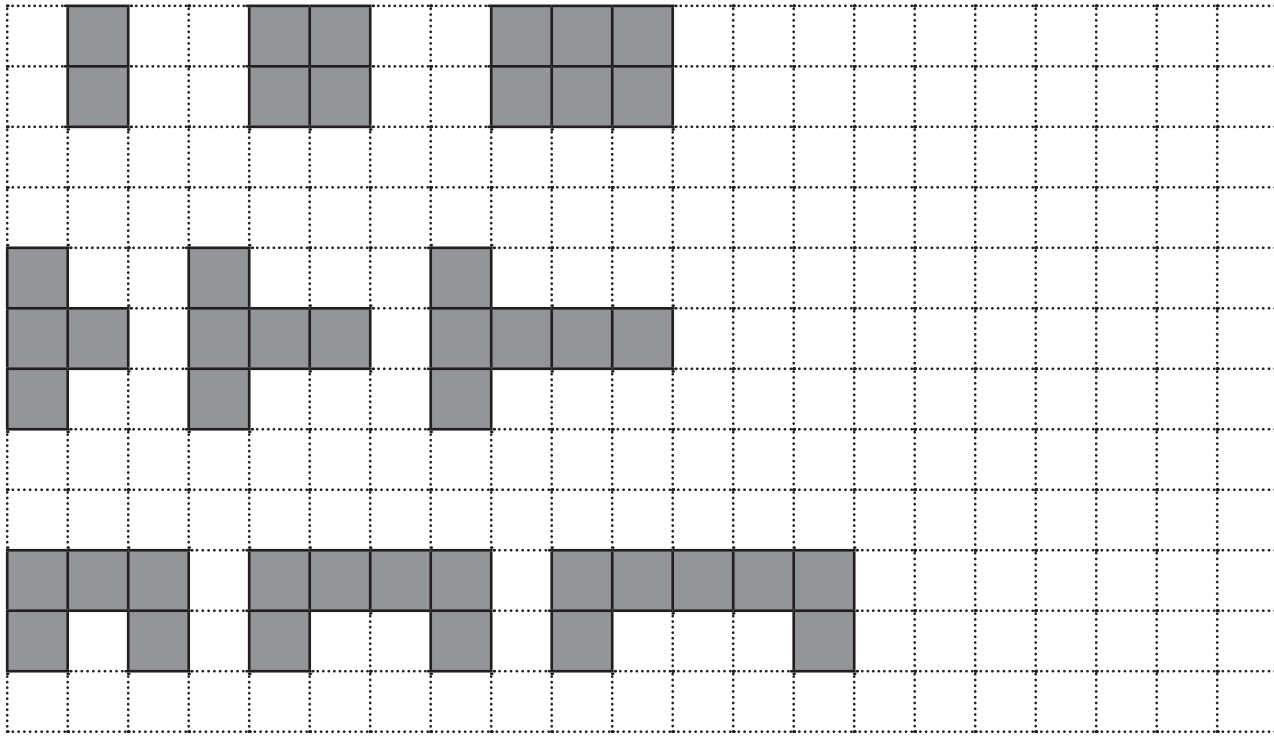
# 3.2 Building patterns with squares

**1** Lachlan made these patterns by shading squares on grid paper.
Draw the next step in each pattern.

**a**

**b**

**c**



**2** For each of the patterns above:

- count the number of shaded squares in each of the first four steps
- complete the matching table below
- predict the number of squares in step 5.

**a**

| **Step in the pattern** | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| **Number of shaded squares** | | | | | |

**b**

| **Step in the pattern** | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| **Number of shaded squares** | | | | | |

**c**

| **Step in the pattern** | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| **Number of shaded squares** | | | | | |

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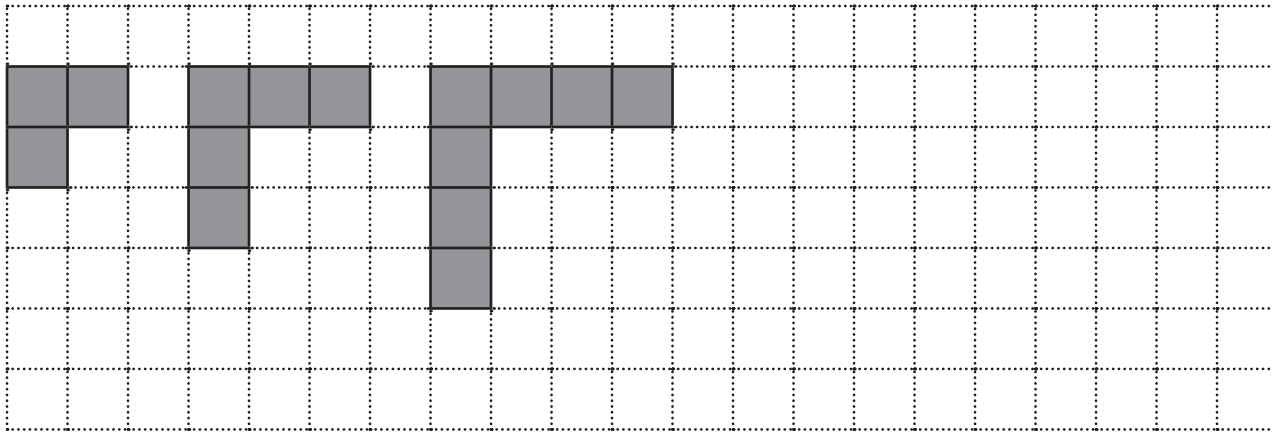
# 3.2 Building patterns with squares

**CONTINUED**

**3** **a** Draw the next step in this pattern.

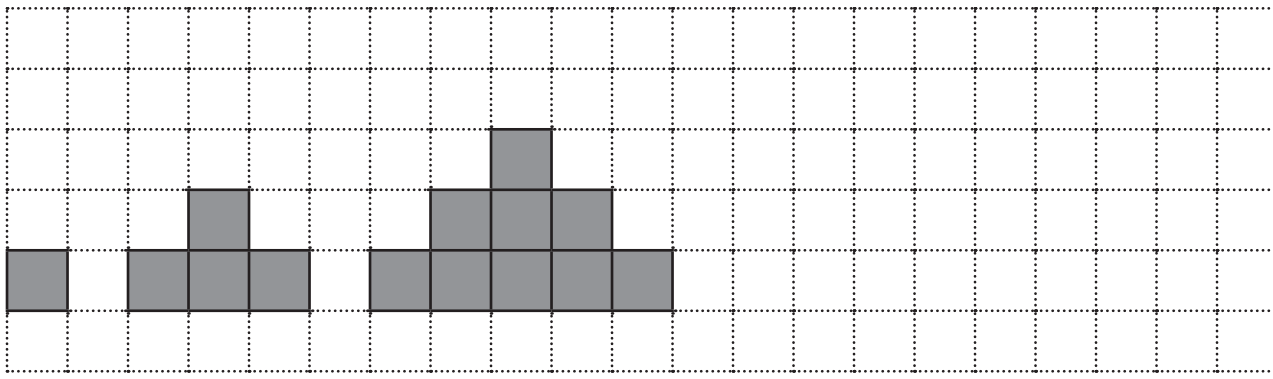


**b** Count the number of shaded squares in each of the first four steps in the pattern above. Write your answers in this table.

| **Step in the pattern** | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| **Number of shaded squares** | | | | | | |

**c** Predict the number of shaded squares in step 5 and step 6.

**4** **a** Draw the next step in this pattern.



**b** Count the number of shaded squares in each of the first four steps in the pattern above. Write your answers in this table.

| **Step in the pattern** | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| **Number of shaded squares** | | | | | | |

**c** Predict the number of shaded squares in step 5 and step 6.

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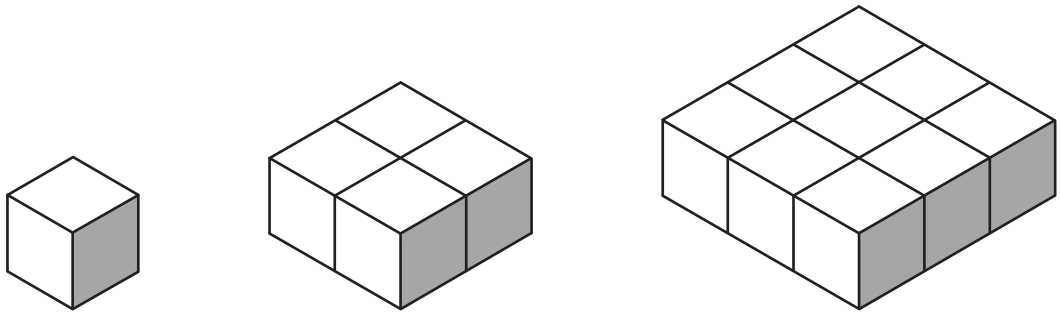
# 3.3 Building patterns with cubes

Annaka built these patterns using cubes.
For each pattern:

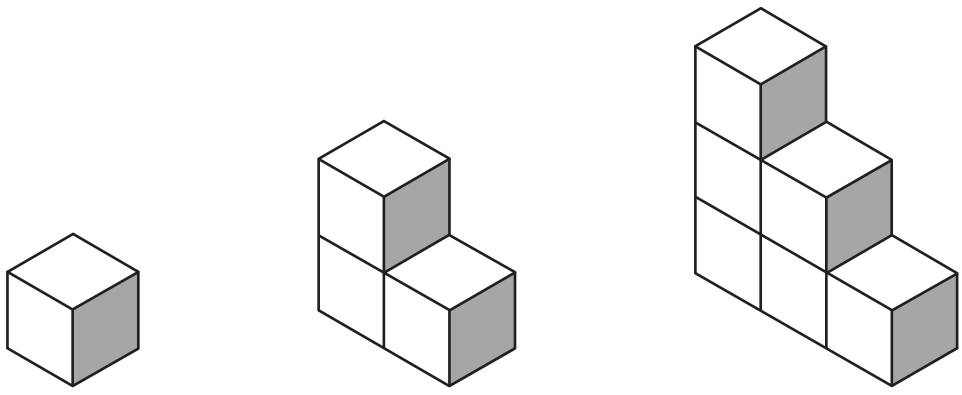
- **a** build the first 3 steps, as shown
- **b** use cubes to build steps 4 and 5.
- **c** count the number of cubes in each step and complete the table of values.

**1**



| Step in the pattern | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| Number of cubes | | | | | |

**2**



| Step in the pattern | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| Number of cubes | | | | | |

**3** Look at the pattern of numbers in the table of question 2.
Predict how many cubes are needed to build step 6 in this pattern. _____

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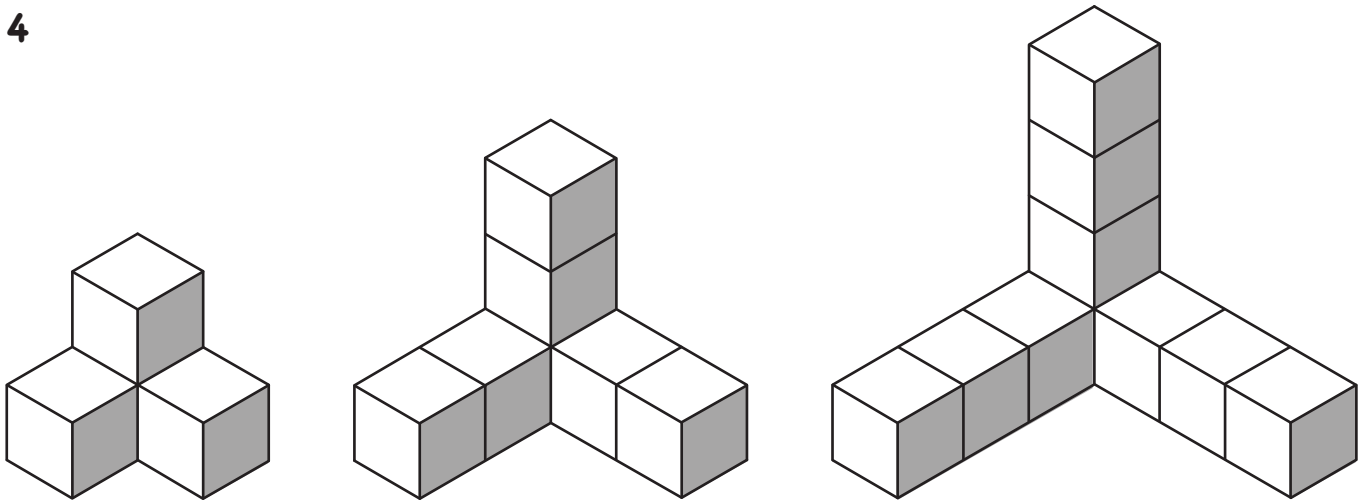
# 3.3 Building patterns with cubes

**CONTINUED**

**4**

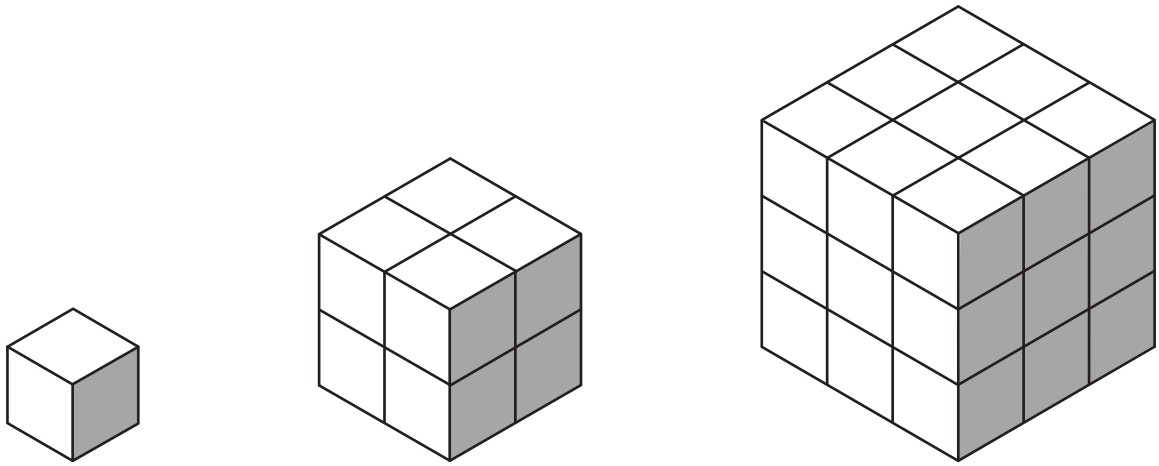


| Step in the pattern | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| Number of cubes | | | | | |

**5** Predict the number of cubes needed to build the next two steps in question 4.

**a** Step 6 _____ **b** Step 7 _____

**6**



| Step in the pattern | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| Number of cubes | | | | | |

**7** Predict the number of cubes needed to build the next step (Step 6) in question 6.

_____

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# 3.4 Describing geometric patterns

In these activities, you will be looking at patterns made using toothpicks.
You will write a rule to describe each pattern.

**1**

**a** Count the number of toothpicks needed to build each step in this pattern. Write your answers in the table and continue the pattern.

| **Number of triangles** | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| **Number of toothpicks** | | | | | | | | |

**b** Complete this sentence to describe the number of toothpicks needed to build a given number of triangles.

*The number of toothpicks is equal to* _____ *the number of triangles.*

**c** How many toothpicks will be needed to build:

**(i)** 20 triangles? _____ **(ii)** 50 triangles? _____

**2**

**a** Count the number of toothpicks needed to build each step in this pattern. Write your answers in the table and continue the pattern.

| **Number of squares** | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| **Number of toothpicks** | | | | | | | | |

**b** Complete this sentence to describe the number of toothpicks needed to build a given number of squares.

*The number of toothpicks is equal to 3 times the number of triangles then*

_____.

**c** How many toothpicks will be needed to build:

**(i)** 20 squares? _____ **(ii)** 50 squares? _____

# 3.4 Describing geometric patterns

**CONTINUED**

**3**

**a** Count the number of toothpicks needed to build each step in this pattern. Write your answers in the table and continue the pattern.

| **Number of triangles** | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| **Number of toothpicks** | | | | | | | | |

**b** Complete this sentence to describe the number of toothpicks needed to build a given number of triangles.

*The number of toothpicks is equal to* _____ *the number of triangles, then add 1.*

**c** How many toothpicks will be needed to build:

**(i)** 25 triangles? _____ **(ii)** 100 triangles? _____

**4**

**a** Count the number of toothpicks needed to build each step in this pattern. Write your answers in the table and continue the pattern.

| **Number of hexagons** | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| **Number of toothpicks** | | | | | | | | |

**b** Complete this sentence to describe the number of toothpicks needed to build a given number of hexagons.

*The number of toothpicks is equal to* _____ *the number of hexagons, then* _____.

**c** How many toothpicks will be needed to build:

**(i)** 30 hexagons? _____ **(ii)** 45 hexagons? _____

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# 3.5 Using pattern rules

**1** Jawain used toothpicks to build this pattern of squares.

In this pattern, the number of toothpicks is equal to 4 times the number of squares.

How many toothpicks are needed to build:

**a** 5 squares? *Number of toothpicks = 4 × 5 = 20*

**b** 12 squares? _____

**To answer the following questions, you must use the given rule.**

**2** The number of toothpicks is equal to 3 times the number of triangles.
How many toothpicks are needed to build:

**a** 1 triangle? _____ **b** 3 triangles? _____

**c** 5 triangles? _____ **d** 12 triangles? _____

**3** The number of toothpicks is equal to 6 times the number of hexagons.
How many toothpicks are needed to build:

**a** 1 hexagon? _____ **b** 4 hexagons? _____

**c** 6 hexagons? _____ **d** 10 hexagons? _____

**4** The number of toothpicks is equal to 2 times the number of triangles, then **add 1**.

How many toothpicks are needed to build:

**a** 2 triangles? _____ **b** 5 triangles? _____

**c** 8 triangles? _____ **d** 12 triangles? _____

**5** The number of toothpicks is equal to 4 times the number of triangles, then **subtract 1**.

How many toothpicks are needed to build:

**a** 3 triangles? _____ **b** 5 triangles? _____

**c** 9 triangles? _____ **d** 15 triangles? _____

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# 3.6 Describing number patterns

Perrine wrote the first five terms in a number pattern: **5, 8, 11, 14, 17**
What is the next term in this pattern?

This pattern can be described in different ways:

- ***Each number is 3 more than the previous number.***
- ***Starting with 5, add 3 to the previous number.***

The next term will be **17 + 3 = 20.**

**1** The following number patterns were formed by **adding**.
Write the next four terms in the pattern and describe the pattern in words.

**a** 3, 5, 7, 9, *11*, *13*, *15*, *17* — *Start with 3, add 2*

**b** 1, 4, 7, 10, ____, ____, ____, ____ — ____

**c** 5, 10, 15, 20, ____, ____, ____, ____ — ____

**d** 6, 10, 14, 18, ____, ____, ____, ____ — ____

**e** 2, 7, 12, 17, ____, ____, ____, ____ — ____

**f** 4, 13, 22, 31, ____, ____, ____, ____ — ____

**2** The following number patterns were formed by **subtracting**.
Write the next four terms in the pattern and describe the pattern in words.

**a** 100, 98, 96, 94, *92*, *90*, *88*, *86* — *Start with 100, subtract 2*

**b** 10, 9, 8, 7, ____, ____, ____, ____ — ____

**c** 20, 18, 16, 14, ____, ____, ____, ____ — ____

**d** 100, 95, 90, 85, ____, ____, ____, ____ — ____

**e** 36, 32, 28, 24, ____, ____, ____, ____ — ____

**f** 50, 44, 38, 32, ____, ____, ____, ____ — ____

**3** The following number patterns were formed by **multiplying**.
Write the next three terms in the pattern and describe the pattern in words.

**a** 1, 2, 4, 8, *16*, *32*, *64* — *Start with 1, multiply by 2*

**b** 3, 6, 12, 24, ____, ____, ____ — ____

**c** 1, 3, 9, 27, ____, ____, ____ — ____

**d** 5, 10, 20, 40, ____, ____, ____ — ____

**e** 2, 10, 50, 250, ____, ____, ____ — ____

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# 3.6 Describing number patterns

**CONTINUED**

**4** The following number patterns were formed by **dividing**.
Write the next two terms in the pattern and describe the pattern in words.

**a** 64, 32, 16, 8, *4*, *2* — *Start with 64, divide by 2*

**b** 96, 48, 24, 12, _____, _____ — _____

**c** 243, 81, 27, 9, _____, _____ — _____

**d** 100 000, 10 000, 1 000, 100, _____, _____ — _____

**e** 1 215, 405, 135, 45, _____, _____ — _____

**5** Use the starting number and the rule to write the next three terms in each pattern.

**a** Starting with 3, add 2.

*3*, *5*, *7*, *9*

**b** Starting with 2, add 3.

_____, _____, _____, _____

**c** Starting with 4, add 5.

_____, _____, _____, _____

**d** Starting with 7, add 4.

_____, _____, _____, _____

**e** Starting with 5, add 9.

_____, _____, _____, _____

**f** Starting with 12, subtract 2.

_____, _____, _____, _____

**g** Starting with 50, subtract 5.

_____, _____, _____, _____

**h** Starting with 36, subtract 6.

_____, _____, _____, _____

**i** Starting with 3, multiply by 2.

_____, _____, _____, _____

**j** Starting with 10, multiply by 2.

_____, _____, _____, _____

**k** Starting with 5, multiply by 3.

_____, _____, _____, _____

**l** Starting with 9, multiply by 4.

_____, _____, _____, _____

**m** Starting with 32, divide by 2.

_____, _____, _____, _____

**n** Starting with 1 000, divide by 5.

_____, _____, _____, _____

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# Teacher's notes

## OBJECTIVES

- Generate, describe and record geometric patterns using a variety of strategies.
- Complete a table of values for a given pattern.
- Extend a number pattern and find a higher term, given the first four terms.
- Use letters to represent numbers when describing patterns.
- Translate between words and algebraic symbols and between algebraic symbols and words.

## PRIOR KNOWLEDGE

Students should be able to perform the basic operations with whole numbers and fractions using mental arithmetic or a calculator.

## BACKGROUND

These activity sheets provide a variety of geometric patterns so that students can see the development of algebra in a variety of situations. It is important to develop an understanding of the use of letters as algebraic symbols for variable **numbers of objects** rather than for the objects themselves. The practice of using the first letter of the name of an object as a symbol for the object (e.g. 'a' represents an apple) leads to students developing incorrect concepts and should be avoided.

## STARTER ACTIVITY

Ask students to count forwards or backwards by ones, twos, threes, fours, fives or tens, using different starting values, on and off the multiple.

Alternatively, count aloud the first five terms then stop and ask students: *What is the next term in this pattern*?

## ACTIVITY SHEETS

### 4.1 Writing rules using symbols A

This activity sheet follows a process of building a geometric pattern, completing a table of values, describing the pattern in words and then writing an algebraic equation that describes the pattern.

Patterns are built using toothpicks.

**Answers**

**1** **b** 8, 10, 12

**2** **b** 3, 4, 5, 6, 7
**c** The number of toothpicks needed is equal to the step number plus **one.**
**d** $T = S + 1$

**3** **b** 5, 7, 9, 11, 13
**c** The number of toothpicks needed is **two** times the step number plus one.
**d** $T = 2 \times S + 1$ **e** 25

**4** **b** 10, 14, 18, 22, 26
**c** The number of toothpicks needed is **four** times the step number plus two.
**d** $T = 4 \times S + 2$
**e** 82

### 4.2 Writing rules using symbols B

This activity sheet follows a process of building a geometric pattern, completing a table of values, describing the pattern in words and then writing an algebraic equation that describes the pattern.

Patterns are built using shaded squares on a grid.

**Answers**

**1** **b** 8, 10, 12

**2** **b** 4, 5, 6, 7, 8
**c** The number of shaded squares is equal to the step number plus **two**.
**d** $N = S + 2$

**3** **b** 6, 7, 8, 9, 10
**c** The number of shaded squares is equal to the step number **plus four**.
**d** $N = S + 4$
**e** 16

**4** **b** 5, 7, 9, 11, 13
**c** The number of shaded squares is equal to **two** times the step number plus **one**.
**d** $N = 2 \times S + 1$
**e** 41

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# Teacher's notes

CONTINUED

### 4.3 Using pronumerals

In this activity sheet, students write an equation to express a pattern rule that is given as a sentence. They are also asked to write sentences to translate algebraic equations.

**Answers**

**1** **a** $T = 3 \times S$ **b** $T = 4 \times S$ **c** $T = S + 1$
**d** $T = S + 4$ **e** $T = S - 1$ **f** $T = 3 \times S - 2$
**g** $T = 3 \times S + 1$ **h** $T = 2 \times S + 3$
**i** $T = 2 \times S - 1$ **j** $T = 3 \times S - 5$

**2** **b** The number of toothpicks equals the step number plus two.
**c** The number of toothpicks equals the step number plus three.
**d** The number of toothpicks is two times the step number plus one.
**e** The number of toothpicks is two times the step number minus three.
**f** The number of toothpicks is four times the step number minus two.

**3** **a** $N = 3 \times S$ **b** $N = 4 \times S$ **c** $N = S + 3$
**d** $N = 3 \times S - 4$ **e** $N = 2 \times S + 3$
**f** $N = 4 \times S - 1$

### 4.4 Finding number patterns

On the first page of this activity sheet, students are given equations which describe a pattern. They are asked to find the first four terms in each pattern. On the second page, they find the first ten terms in various patterns described by algebraic equations. They are asked to write their answers in tables of values. Some answers include fractions or negative numbers.

**Answers**

**1** **a** 4, 5, 6, 7 **b** -1, 0, 1, 2 **c** 5, 10, 15, 20
**d** 3, 5, 7, 9 **e** 0.5, 1, 1.5, 2

**2** **a** 6, 7, 8, 9, 10, 11, 12, 13, 14, 15
**b** -2, -1, 0, 1, 2, 3, 4, 5, 6, 7
**c** 4, 8, 12, 16, 20, 24, 28, 32, 36, 40
**d** 5, 7, 9, 11, 13, 15, 17, 19, 21, 23
**e** 7, 10, 13, 16, 19, 22, 25, 28, 31, 34
**f** 1, 3, 5, 7, 9, 11, 13, 15, 17, 19
**g** 2, 7, 12, 17, 22, 27, 32, 37, 42, 47
**h** 14, 13, 12, 11, 10, 9, 8, 7, 6, 5

## EXTENSION IDEAS

Play *Guess my Rule* games where you present students with geometric patterns and ask students to describe the pattern rule in words and using algebraic symbols. Students could also work in small groups to make patterns, exchange them with another group and describe them.

Ask students to represent the values from a table on a number grid and describe the pattern formed by the points on the graph.

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# 4.1 Writing rules using symbols A

A quick and simple way to write general rules for patterns is to use symbols. The symbols are called **pronumerals**. They are used in place of numbers. Letters of the alphabet are usually used as pronumerals.

Toothpicks were used to build these patterns.

**1**

**a** Draw the next step in this pattern.

**b** Write the number of toothpicks needed to build each step in this table.

| **Step in the pattern (*S*)** | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| **Number of toothpicks (*T*)** | 2 | 4 | 6 | | | |

The number of toothpicks needed is two times the step number.
We can write this rule using pronumerals:

$T = 2 \times S$

***T*** stands for the number of toothpicks.
***S*** stands for the step number.

**2**

**a** Draw the next step in this pattern.

**b** Write the number of toothpicks needed to build each step in this table.

| **Step in the pattern (*S*)** | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| **Number of toothpicks (*T*)** | 2 | | | | | |

**c** Complete this rule to describe the number of toothpicks needed to build each step.

*The number of toothpicks needed is equal to the step number plus* _____.

**d** Write this rule using pronumerals. _____

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# 4.1 Writing rules using symbols A

**CONTINUED**

**3**

**a** Draw the next step in this pattern.

**b** Complete this table.

| **Step in the pattern ($S$)** | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| **Number of toothpicks ($T$)** | 3 | | | | | |

**c** Complete this rule to describe the number of toothpicks needed to build each step.

*The number of toothpicks needed is _____ times the step number plus 1.*

**d** Write this rule using pronumerals. _____

**e** How many toothpicks will be needed to build step 12? _____

**4**

**a** Draw the next step in this pattern.

**b** Complete this table.

| **Step in the pattern ($S$)** | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| **Number of toothpicks ($T$)** | 6 | | | | | |

**c** Complete this rule to describe the number of toothpicks needed to build each step.

*The number of toothpicks needed is _____ times the step number plus 2.*

**d** Write this rule using pronumerals. _____

**e** How many toothpicks will be needed to build step 20? _____

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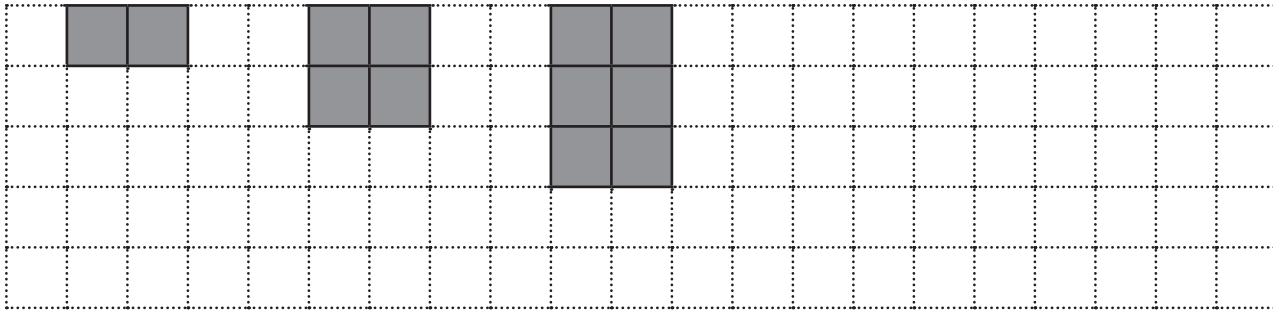
# 4.2 Writing rules using symbols B

A quick and simple way to write general rules for patterns is to use symbols. The symbols are called **pronumerals**. They are used in place of numbers. Letters of the alphabet are usually used as pronumerals.

These patterns were made by shading squares on grid paper.

**1**



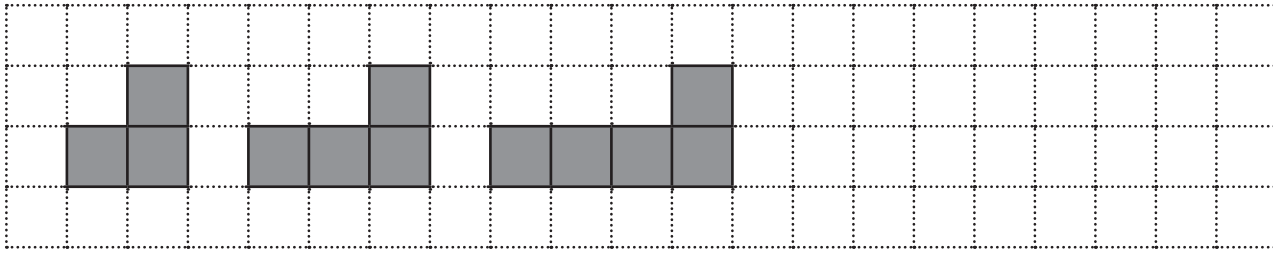
**a** Draw the next step in this pattern.

**b** Write the number of shaded squares in each step in this table.

| **Step in the pattern (*S*)** | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| **Number of shaded squares (*N*)** | 2 | 4 | 6 | | | |

The number of shaded squares is two times the step number.
We can write this rule using pronumerals: $N = 2 \times S$.
$N$ stands for the number of shaded squares. $S$ stands for the step number.

**2**



**a** Draw the next step in this pattern.

**b** Write the number of shaded squares in each step in this table.

| **Step in the pattern (*S*)** | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| **Number of shaded squares (*N*)** | 3 | | | | | |

**c** Complete this rule to describe the pattern.
*The number of shaded squares is equal to the step number plus* _____.

**d** Write this rule using pronumerals. _____

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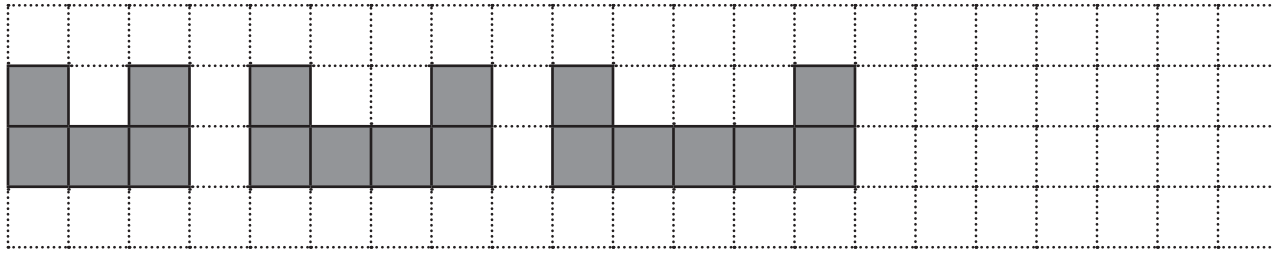
# 4.2 Writing rules using symbols B

**CONTINUED**

**3**



**a** Draw the next step in this pattern.

**b** Write the number of shaded squares in each step in this table.

| **Step in the pattern (*S*)** | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| **Number of shaded squares (*N*)** | 5 | | | | | |

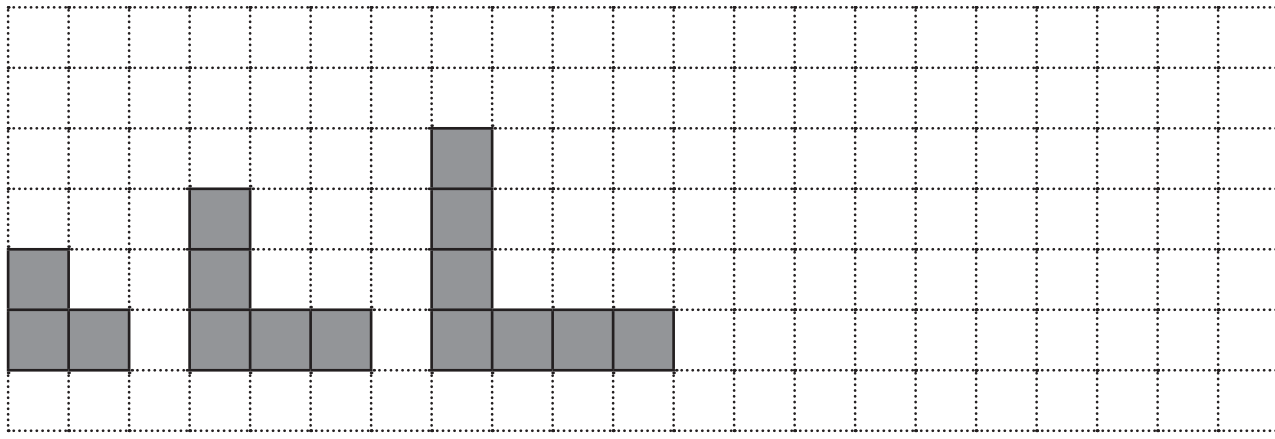
**c** Complete this rule to describe the pattern.

*The number of shaded squares is equal to the step number* _____.

**d** Write this rule using pronumerals. _____

**e** How many squares will be shaded in step 12? _____

**4**



**a** Draw the next step in this pattern.

**b** Write the number of shaded squares in each step in this table.

| **Step in the pattern (*S*)** | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| **Number of shaded squares (*N*)** | 3 | | | | | |

**c** Complete this rule to describe the pattern.

*The number of shaded squares is* _____ *times the step number plus* _____ .

**d** Write this rule using pronumerals. _____

**e** How many squares will be shaded in step 20? _____

# 4.3 Using pronumerals

Joshua built a pattern using toothpicks. The number of toothpicks used was three times the step number. He wrote the rule $T = 3 \times S$ to describe the pattern.

$T$ stands for the number of toothpicks. $S$ stands for the step number in the pattern.

**1** Use the pronumerals $T$ and $S$ to write the following pattern rules.
$T$ stands for the number of toothpicks.
$S$ stands for the step number in the pattern.

**a** The number of toothpicks is three times the step number.

*$T = 3 \times S$*

**b** The number of toothpicks is four times the step number.

**c** The number of toothpicks equals the step number plus 1.

**d** The number of toothpicks equals the step number plus 4.

**e** The number of toothpicks equals the step number minus 1.

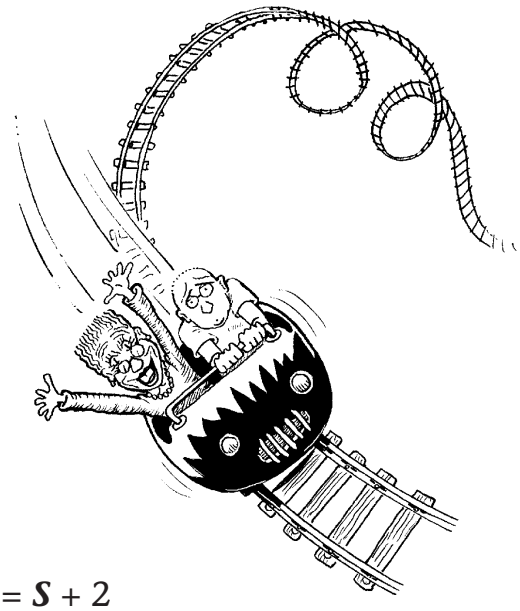
**f** The number of toothpicks is three times the step number minus 2.

**g** The number of toothpicks is three times the step number plus 1.

**h** The number of toothpicks is two times the step number plus 3.

**i** The number of toothpicks is two times the step number minus 1.

**j** The number of toothpicks is three times the step number minus 5.



**2** Use words to write these pattern rules.
$T$ stands for the number of toothpicks.
$S$ stands for the step number.

**a** $T = 2 \times S$

*The number of toothpicks is two times the step number.*

**b** $T = S + 2$

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# 4.3 Using pronumerals

**CONTINUED**

**c** $T = S + 3$

**d** $T = 2 \times S + 1$

**e** $T = 2 \times S - 3$

**f** $T = 4 \times S - 2$

**3** Use the pronumerals $N$ and $S$ to write the following pattern rules.
$N$ stands for the number of shaded squares.
$S$ stands for the step number in the pattern.

**a** The number of squares is equal to three times the step number.

$N = 3 \times S$

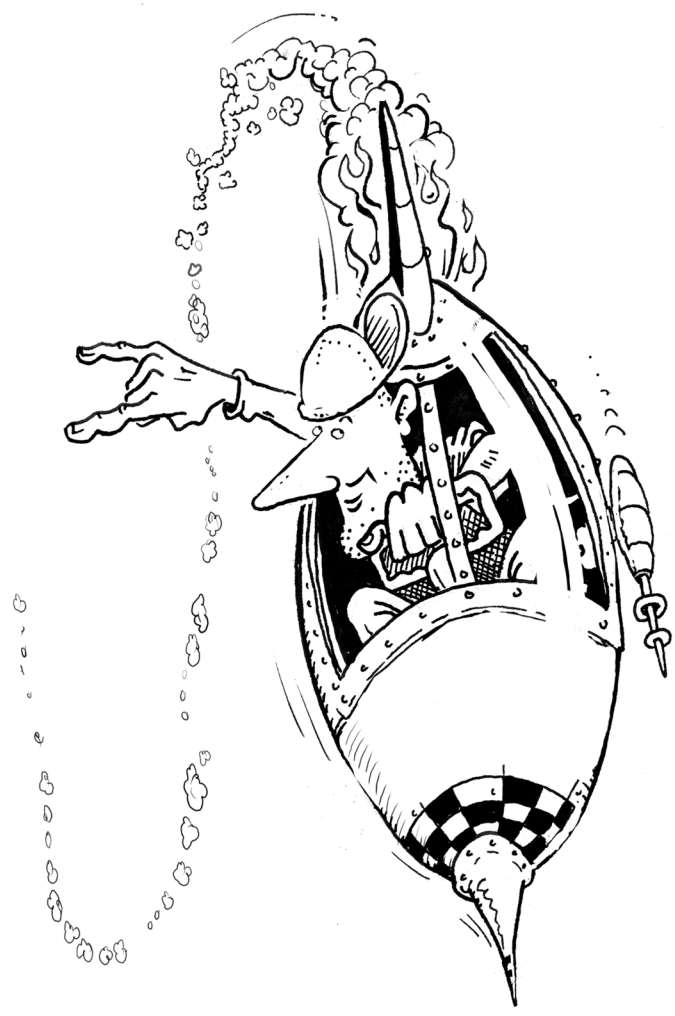
**b** The number of squares is equal to four times the step number.

**c** The number of squares is equal to the step number plus 3.

**d** The number of squares is equal to three times the step number minus 4.

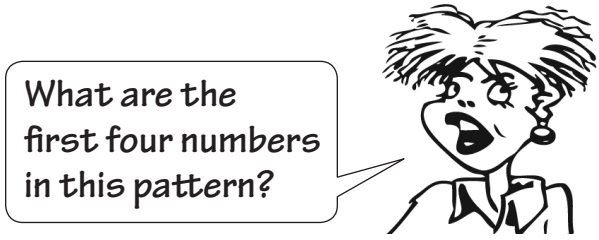
**e** The number of squares is equal to two times the step number plus 3.

**f** The number of squares is equal to four times the step number minus 1.



# 4.4 Finding number patterns

Marianne said the rule for a number pattern was $T = n + 4$.
The pronumeral $T$ stands for the value of the term in the pattern.
The pronumeral $n$ stands for the term number.
She asked Derek:




We often use a rule with pronumerals to describe a number pattern.

The pronumerals stand for numbers.

To find the first four terms, Derek needs to replace $n$ with 1, then 2, 3, and 4.

**EXAMPLE**

$T = n + 4$

put $n = 1$, $T = 1 + 4$
$= 5$

put $n = 2$, $T = 2 + 4$
$= 6$

put $n = 3$, $T = 3 + 4$
$= 7$

put $n = 4$, $T = 4 + 4$
$= 8$

The first four terms in this pattern are **5**, **6**, **7** and **8**.

**1** Find the first four terms in the number pattern described by each rule.

**a** $T = n + 3$ put $n = 1$, T = _____ put $n = 2$, T = _____
put $n = 3$, T = _____ put $n = 4$, T = _____
The first four terms in this pattern are: _____, _____, _____, _____

**b** $T = n - 2$ put $n = 1$, T = _____ put $n = 2$, T = _____
put $n = 3$, T = _____ put $n = 4$, T = _____
The first four terms in this pattern are: _____, _____, _____, _____

**c** $T = 5 \times n$ put $n = 1$, T = _____ put $n = 2$, T = _____
put $n = 3$, T = _____ put $n = 4$, T = _____
The first four terms in this pattern are: _____, _____, _____, _____

**d** $T = 2 \times n + 1$ put $n = 1$, T = _____ put $n = 2$, T = _____
put $n = 3$, T = _____ put $n = 4$, T = _____
The first four terms in this pattern are: _____, _____, _____, _____

**e** $T = n \div 2$ put $n = 1$, T = _____ put $n = 2$, T = _____
put $n = 3$, T = _____ put $n = 4$, T = _____
The first four terms in this pattern are: _____, _____, _____, _____

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# 4.4 Finding number patterns

**CONTINUED**

**2** Find the first ten terms in the number patterns described by the following rules. Write your answers under the term number in the table provided.

**a** $T = n + 5$

| $n$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| $T$ | 6 | 7 | 8 | | | | | | | |

**b** $T = n - 3$

| $n$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| $T$ | | | | | | | | | | |

**c** $T = 4 \times n$

| $n$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| $T$ | | | | | | | | | | |

**d** $T = 2 \times n + 3$

| $n$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| $T$ | | | | | | | | | | |

**e** $T = 3 \times n + 4$

| $n$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| $T$ | | | | | | | | | | |

**f** $T = 2 \times n - 1$

| $n$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| $T$ | | | | | | | | | | |

**g** $T = 5 \times n - 3$

| $n$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| $T$ | | | | | | | | | | |

**h** $T = 15 - n$

| $n$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| $T$ | | | | | | | | | | |

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# Teacher's notes

## OBJECTIVES

- Write algebraic expressions to describe various situations.
- Recognise equivalent algebraic expressions.
- Replace a pronumeral with a given value and evaluate algebraic expressions.
- Recognise like algebraic terms.
- Add and subtract like terms to simplify algebraic expressions.
- Simplify algebraic expressions that involve multiplication or division.

## PRIOR KNOWLEDGE

Students should understand the use of a pronumeral to stand for an unknown number. They should be able to perform the basic operations with whole numbers, fractions and decimals using mental arithmetic or a calculator.

## BACKGROUND

Some activity sheets feature or refer to concrete materials. Some students may find it helpful to use materials or diagrams as they work through these activity sheets, especially if they have not fully understood the concept of a pronumeral standing for an unknown variable number.

## STARTER ACTIVITY

Give students the first four terms in a selection of number patterns, such as 2, 4, 6, 8 and 1, 5, 9, 13. Ask them to continue each pattern, describe the pattern in words and write an algebraic rule to describe the pattern.

## ACTIVITY SHEETS

### 5.1 Writing expressions

In this activity sheet, students write algebraic expressions for the number of counters in a variety of situations, involving an unknown variable. If concrete materials are required, toothpicks, cubes or paperclips could be used in place of counters.

**Answers**

**1** **a** $N + 3$ **b** $N + 4$ **c** $N + 5$ **d** $N + 8$
**e** $2 \times N$ **f** $3 \times N$ **g** $2 \times N + 3$ **h** $2 \times N + 5$
**i** $3 \times N + 2$ **j** $3 \times N + 4$ **k** $4 \times N + 7$
**l** $5 \times N + 6$

**2** **a** $2 \times A$ **b** $A + 2$ **c** $A + 5$ **d** $2 \times A + 4$
**e** $2 \times A + 9$ **f** $2 \times A + 8$ **g** $3 \times A + 7$
**h** $4 \times A + 3$

**3** **a** $N + A$ **b** $2 \times N + A$ **c** $N + 2 \times A$
**d** $3 \times N + A$ **e** $2 \times N + 2 \times A$ **f** $2 \times N + 2 \times A$

### 5.2 Simplifying algebraic expressions

This activity sheet introduces some of the notation of algebra. Students learn to write simple expressions for statements like 3 lots of $N$, $3 \times N$, $a + a + a + a$ and $a \times b$.

**Answers**

**1** **a** $3y$ **b** $5A$ **c** $4N$ **d** $2m$ **e** $3a$ **f** $4m$
**g** $6y$ **h** $7A$ **i** $2x$ **j** $5y$ **k** $8m$ **l** $5a$
**m** $my$ **n** $xy$

**2** **a** $3x$ **b** $4y$ **c** $2a + b$ **d** $a + 3b$ **e** $5m$
**f** $6N$ **g** $ab$ **h** $km$ **i** $3a + 2b$ **j** $2A + 2N$
**k** $4ab$ **l** $3km$ **m** $abc$ **n** $kmn$ **o** $2abc$
**p** $8kmn$

### 5.3 Evaluating expressions

In this activity sheet, students learn to replace a pronumeral with a given value and evaluate simple algebraic expressions.

**Answers**

**1** **a** 4 **b** 7 **c** 8 **d** 2 **e** 1 **f** 3 **g** 9 **h** 15
**i** 6 **j** 7 **k** 11 **l** 5 **m** 13 **n** 2 **o** 19
**p** 13

**2** **a** 11 **b** 7 **c** 7 **d** 2 **e** 2 **f** 6 **g** 15
**h** 10 **i** 24 **j** 7 **k** 3 **l** 11 **m** 13 **n** 14
**o** 3 **p** 10 **q** 11 **r** 14 **s** 12 **t** 30 **u** 16
**v** 40 **w** 160 **x** 34

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# Teacher's notes

CONTINUED

## 5.4 Adding and subtracting algebraic terms

This activity sheet starts with the recognition of like terms, before proceeding to addition and subtraction. Emphasise that only like terms can be added and subtracted. This is a very important idea and students must understand that they are simplifying the expressions by 'collecting like terms'. Some students may find it helpful to use concrete materials or diagrams as shown in the diagrams on the first page.

**Answers**

**1** **a** $5a$, $8a$ **b** $3a$, $7a$ **c** $3y$, $7y$ **d** $3x$, $x$ **e** $2m$, $-3m$ **f** $8k$, $3k$ **g** $-2x$, $6x$ **h** $5n$, $-8n$ **i** $4k$, $8k$ **j** $3x$, $5x$ **k** $4b$, $8b$ **l** $4k$, $-2k$ **m** $-5$, $7$ **n** $4mn$, $7mn$ **o** $2ab$, $11ab$

**2** **a** $7a$ **b** $7n$ **c** $11k$ **d** $13m$ **e** $19y$ **f** $10x$ **g** $22k$ **h** $2m$ **i** $16y$ **j** $12a$ **k** $15g$ **l** $21x$

**3** **a** $4n$ **b** $4a$ **c** $2m$ **d** $5x$ **e** $m$ **f** $4y$ **g** $7a$ **h** $8n$ **i** $0$ **j** $y$ **k** $7n$ **l** $14m$

**4** **a** $11m + 2a$ **b** $7n + 4$ **c** $4n + 12$ **d** $8x + 6y$ **e** $5a + 6$ **f** $7a$ **g** $20k + 3m$ **h** $10m$ **i** $3n + 8$ **j** $5y + 3$ **k** $6x$ **l** $3m - 7k$ **m** $10xy$ **n** $4ab$ **o** $13k - 5m$ **p** $7m - 7a$

**5** **a** $8n + 6$ **b** $5a + 11$ **c** $11x + 3y$ **d** $7a + 7b$ **e** $10k + 10m$ **f** $8x + 16y$ **g** $5n + 10$ **h** $2m + 2$ **i** $4a + 1$ **j** $10x + 2y$ **k** $2k + m$ **l** $4ab + 5$

## 5.5 Multiplying with pronumerals

In this activity sheet, students learn to multiply simple algebraic expressions. Remind students to multiply the numbers first, then multiply the pronumerals. The pronumerals are usually written in alphabetical order, but this is not essential. Accept an answer *2ba* in place of *2ab*, but remind students of the preference for alphabetical order. This activity sheet introduces index notation to simplify $N \times N \times N$, but does **not** include expressions involving grouping symbols.

**Answers**

**1** **a** $4a$ **b** $5y$ **c** $8m$ **d** $ab$ **e** $km$ **f** $3XY$ **g** $6gh$ **h** $abc$ **i** $kmn$

**2** **a** $5 \times m$ **b** $3 \times k$ **c** $x \times y$ **d** $p \times q \times r$ **e** $9 \times a \times b$ **f** $7 \times p \times q$

**3** **a** $6y$ **b** $20n$ **c** $12a$ **d** $40m$ **e** $27k$ **f** $20y$ **g** $28a$ **h** $3k$ **i** $4m$ **j** $12mn$ **k** $28ab$ **l** $16ab$ **m** $20ky$ **n** $45km$ **o** $42pq$ **p** $60a$ **q** $35mn$ **r** $12ky$ **s** $24abc$ **t** $56ab$ **u** $60kmn$ **v** $15abc$

**4** **a** $n^4$ **b** $a^3$ **c** $m^5$ **d** $y^2$ **e** $a^2b^3$ **f** $k^2m^2$ **g** $pq^2$ **h** $a^2b$ **i** $b^2c^2$ **j** $k^3y^2$ **k** $a^3b^2$ **l** $k^2n^3$

**5** **a** $9m^3$ **b** $20a^2$ **c** $16a^2$ **d** $21y^2$ **e** $3d^2$ **f** $10m^3$ **g** $10n^2$ **h** $12a^3$ **i** $48m^3$ **j** $24k^2$ **k** $8p^4$ **l** $80m^2$ **m** $3k^2y$ **n** $5am^2$ **o** $24m^2n$ **p** $2a^2b$ **q** $a^2b^2$ **r** $30k^2y$ **s** $6a^2c^2$ **t** $12m^2n^2$

## 5.6 Division with pronumerals

In the first question of this activity sheet, students write algebraic expressions for some sentences which express a division. They are then shown how to simplify expressions which involve a division. Suggest that students write the divisions as fractions, then cancel common factors, as they would do to simplify a numerical fraction. This activity sheet does **not** include indices.

**Answers**

**1** **a** $N \div 5$ **b** $n \div 4$ **c** $Y \div 6$ **d** $12 \div n$ **e** $30 \div x$

**2** **a** $5a$ **b** $2x$ **c** $9m$ **d** $9k$ **e** 4n **f** $5ab$ **g** $a$ **h** $b$ **i** $\frac{a}{4}$ **j** $2c$ **k** $2$ **l** $4m$ **m** $\frac{a}{2}$ **n** $\frac{m}{4}$ **o** $\frac{y}{5}$ **p** $\frac{1}{3}$ **q** $3a$ **r** $\frac{x}{2}$

# EXTENSION IDEAS

Ask students to check whether a simplified expression is correct by substituting numbers for the pronumerals.

Ask students to write algebraic formulae for finding the areas of common geometric figures, such as a rectangle or a triangle.

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# 5.1 Writing algebraic expressions

**Algebraic expressions** contain at least one pronumeral.
A **pronumeral** is a symbol used in place of a number.
Letters of the alphabet are usually used as pronumerals.

**EXAMPLE**

Pedro was given four plastic cups and some counters.
He was asked to put the **same number** of counters in each cup.
He had 5 counters left over.

**Write an algebraic expression for the *total number of counters*.**

The number of counters in each cup = $N$
4 cups of counters + 5 counters
= 4 lots of $N$ + 5
= $4 \times N + 5$
= $4N + 5$

**1** Write an algebraic expression for the **total number of counters** shown in each diagram. Use $N$ for the number of counters in each cup.

**a** _____

**b** _____

**c** _____

**d** _____

**e** _____

**f** _____

**g** _____

**h** _____

**i** _____

**j** _____

**k** _____

**l** _____

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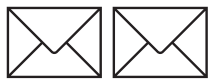
# 5.1 Writing algebraic expressions

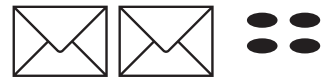
**CONTINUED**

**2** Kathy was given some envelopes and some counters.
She was asked to put the **same number** of counters in each envelope.
Write an algebraic expression for the **total number of counters** shown in each diagram.
Use ***A*** for the number of counters in each envelope.

**a**  _____

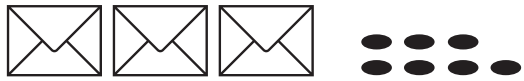
**b** _____

**c** _____

**d**  _____

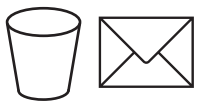
**e** _____

**f** _____

**g**  _____

**h** _____

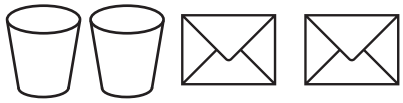
**3** The teacher took some cups from Pedro.
She then took some envelopes from Kathy.
Write an algebraic expression for the total number of counters shown in each diagram.
Use ***N*** for the number of counters in each cup.
Use ***A*** for the number of counters in each envelope.

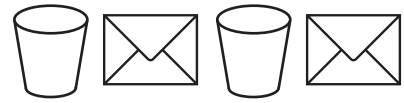
**a**  _____

**b** _____

**c** _____

**d** _____

**e**  _____

**f**  _____

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# 5.2 Simplifying algebraic expressions

In algebra, we prefer to write expressions as simply as possible.
**3 × *k*** or **3 lots of *k*** can be written as **3*k***. We leave out the multiplication sign.
***a* × *b*** can be written as ***ab***.
***N* + *N* + *N* + *N* + *N* = 5 lots of *N*** or **5 × *N* = 5*N***.
Notice that we write the number in front of the pronumeral; that is, 5***N*** **not** ***N***5.

**1** Write the following expressions as simply as possible.

**a** $3 \times y =$ _____

**b** $5 \times A =$ _____

**c** 4 lots of $N =$ _____

**d** $2 \times m =$ _____

**e** $a + a + a =$ _____

**f** $m + m + m + m =$ _____

**g** 6 times $y =$ _____

**h** 7 lots of $A =$ _____

**i** $x + x =$ _____

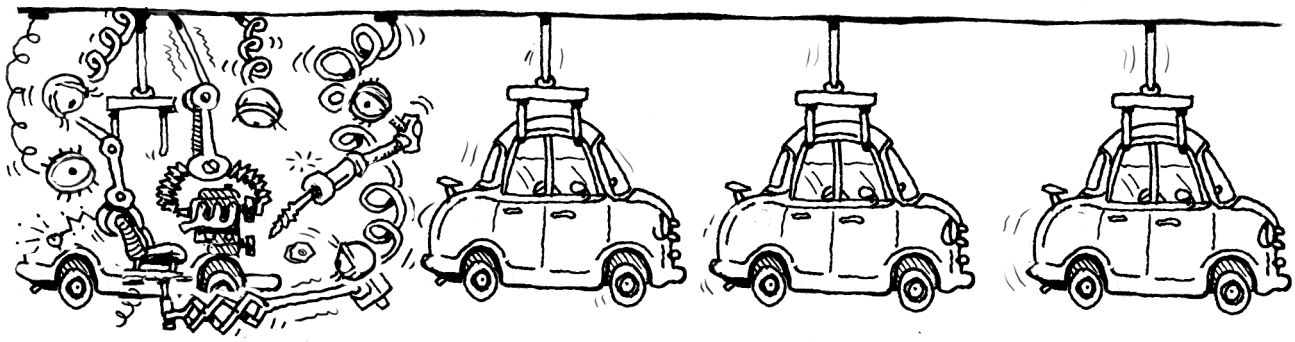
**j** $y + y + y + y + y =$ _____

**k** 8 lots of $m =$ _____

**l** 5 times $a =$ _____

**m** $m \times y =$ _____

**n** $x$ times $y =$ _____



**2** Simplify these expressions.

**a** $x + x + x =$ _____

**b** $y + y + y + y =$ _____

**c** $a + a + b =$ _____

**d** $a + b + b + b =$ _____

**e** $5 \times m =$ _____

**f** $6 \times N =$ _____

**g** $a \times b =$ _____

**h** $k \times m =$ _____

**i** $a + a + a + b + b =$ _____

**j** $A + A + N + N =$ _____

**k** $4 \times a \times b =$ _____

**l** $3 \times k \times m =$ _____

**m** $a \times b \times c =$ _____

**n** $k \times m \times n =$ _____

**o** $2 \times a \times b \times c =$ _____

**p** $8 \times k \times m \times n =$ _____

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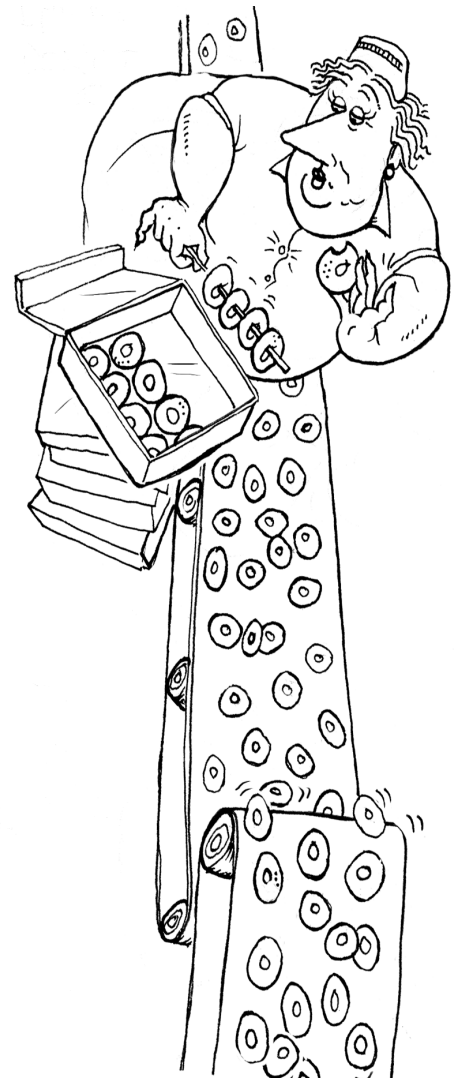
# 5.3 Evaluating expressions

We can find the **value** of an algebraic expression when we know the value of the pronumerals.



**EXAMPLE**

**Find the value of the following expressions when *n* takes the value of 3.**

| $n + 2$ | $n - 3$ | $4n$ |
|---|---|---|
| $= 3 + 2$ | $= 3 - 3$ | $= 4 \times 3$ |
| $= 5$ | $= 0$ | $= 12$ |

**1** Write the value of each expression when $n$ takes the value of 3.

**a** $n + 1 =$ _____ **b** $n + 4 =$ _____

**c** $n + 5 =$ _____ **d** $n - 1 =$ _____

**e** $n - 2 =$ _____ **f** $6 - n =$ _____

**g** $3n =$ _____ **h** $5n =$ _____

**i** $2n =$ _____ **j** $2n + 1 =$ _____

**k** $2n + 5 =$ _____ **l** $2n - 1 =$ _____

**m** $3n + 4 =$ _____ **n** $3n - 7 =$ _____

**o** $4n + 7 =$ _____ **p** $6n - 5 =$ _____

**2** Find the value of each expression when $a = 5$, $b = 2$ and $c = 4$.

**a** $a + 6 =$ _____ **b** $b + 5 =$ _____ **c** $c + 3 =$ _____

**d** $a - 3 =$ _____ **e** $c - 2 =$ _____ **f** $8 - b =$ _____

**g** $3a =$ _____ **h** $5b =$ _____ **i** $6c =$ _____

**j** $a + b =$ _____ **k** $a - b =$ _____ **l** $a + b + c =$ _____

**m** $2a + 3 =$ _____ **n** $5c - 6 =$ _____ **o** $a + b - c =$ _____

**p** $ab =$ _____ **q** $ab + 1 =$ _____ **r** $ab + c =$ _____

**s** $2a + b =$ _____ **t** $3ab =$ _____ **u** $2bc =$ _____

**v** $abc =$ _____ **w** $4abc =$ _____ **x** $3ab + 4 =$ _____

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# 5.4 Adding and subtracting algebraic terms

Here is an algebraic expression with three terms: $3a - 5b + 4$
The three terms are $3a$, $-5b$ and $4$.
Terms are always separated by a plus (+) or minus (–) sign.
$3a$ and $7a$ are called **like terms** because they have the **same pronumerals**.

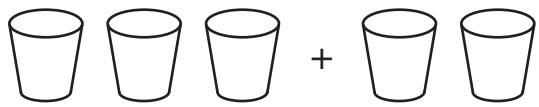
**1** Circle the **like terms** in each list.

**a** (5a) (8a) 4b
**b** 3b 3a 7a
**c** 3y 8x 7y
**d** 2y 3x x
**e** 2m -3m 5k
**f** 8k -7m 3k
**g** -2x -4y 6x
**h** 5n 3 -8n
**i** -5 4k 8k
**j** 3x 4y 5x 6
**k** 2a 4b -6 8b
**l** -3 4k -2k 7m
**m** 4a -5 7 3m
**n** 4mn 3a 7mn
**o** 9a 2ab -3b 11ab

Only **like terms** can be added or subtracted.
This is called **collecting like terms**.

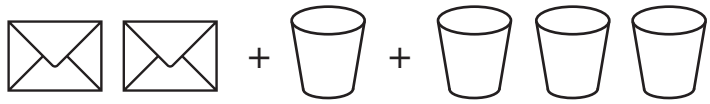
**EXAMPLE**

3n + 2n
= 5n



2n + 4 + n
= 3n + 4

2a + n + 3n
= 2a + 4n



**2** Collect like terms in these expressions.

**a** $4a + 3a =$ _____
**b** $5n + 2n =$ _____
**c** $8k + 3k =$ _____
**d** $9m + 4m =$ _____
**e** $12y + 7y =$ _____
**f** $9x + x =$ _____
**g** $13k + 9k =$ _____
**h** $m + m =$ _____
**i** $5y + 11y =$ _____
**j** $7a + 5a =$ _____
**k** $9g + 6g =$ _____
**l** $8x + 13x =$ _____

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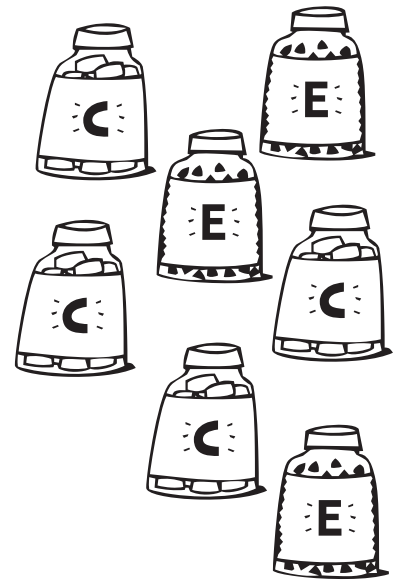
# 5.4 Adding and subtracting algebraic terms

**CONTINUED**




**3** Collect like terms in these expressions.

**a** $9n - 5n =$ _____ **b** $7a - 3a =$ _____

**c** $5m - 3m =$ _____ **d** $11x - 6x =$ _____

**e** $9m - 8m =$ _____ **f** $12y - 8y =$ _____

**g** $8a - a =$ _____ **h** $13n - 5n =$ _____

**i** $7a - 7a =$ _____ **j** $10y - 9y =$ _____

**k** $16n - 9n =$ _____ **l** $15m - m =$ _____

**4** Collect like terms to simplify these expressions.

**a** $6m + 5m + 2a =$ _____ **b** $5n + 2n + 4 =$ _____

**c** $4n + 9 + 3 =$ _____ **d** $8x + 2y + 4y =$ _____

**e** $3a + 6 + 2a =$ _____ **f** $4a + a + 2a =$ _____

**g** $11k + 9k + 3m =$ _____ **h** $6m - 3m + 7m =$ _____

**i** $5n + 8 - 2n =$ _____ **j** $5y + 6 - 3 =$ _____

**k** $16x - 3x - 7x =$ _____ **l** $12m - 7k - 9m =$ _____

**m** $11xy + 5xy - 6xy =$ _____ **n** $7ab - 2ab - ab =$ _____

**o** $8k + 5k - 5m =$ _____ **p** $11m - 7a - 4m =$ _____

**5** Collect like terms to simplify these expressions.

**a** $5n + 2 + 3n + 4 =$ _____ **b** $3a + 5 + 2a + 6 =$ _____

**c** $8x + 2y + 3x + y =$ _____ **d** $6a + 5b + a + 2b =$ _____

**e** $4k + 8m + 6k + 2m =$ _____ **f** $3x + 7y + 9y + 5x =$ _____

**g** $8n + 4 - 3n + 6 =$ _____ **h** $4m + 5 - 2m - 3 =$ _____

**i** $12a + 5 - 4 - 8a =$ _____ **j** $6x + 3y - y + 4x =$ _____

**k** $9k + 2m - 7k - m =$ _____ **l** $5ab + 11 - 6 - ab =$ _____

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# 5.5 Multiplying with pronumerals

In Algebra, the expression $3 \times n$ can be simplified by leaving out the multiplication sign.

$3 \times n$ is written as $3n$.

**1** Simplify the following expressions.

**a** $4 \times a =$ $4a$ **b** $5 \times y =$ _____ **c** $8 \times m =$ _____

**d** $a \times b =$ _____ **e** $k \times m =$ _____ **f** $3 \times X \times Y =$ _____

**g** $6 \times g \times h =$ _____ **h** $a \times b \times c =$ _____ **i** $k \times m \times n =$ _____

**2** Write these expressions in expanded form, using multiplication signs.

**a** $5m =$ $5 \times m$ **b** $3k =$ _____ **c** $xy =$ _____

**d** $pqr =$ _____ **e** $9ab =$ _____ **f** $7pq =$ _____

When you multiply algebraic terms, multiply the numbers first. Then multiply the pronumerals.

The pronumerals are usually written in alphabetical order.

**EXAMPLES**

| | |
|---|---|
| $3 \times 2n$ | $4a \times 3b$ |
| $= 3 \times 2 \times n$ | $= 4 \times a \times 3 \times b$ |
| $= 6n$ | $= 4 \times 3 \times a \times b$ |
| | $= 12ab$ |

**3** Simplify the following expressions.

**a** $2 \times 3y =$ _____ **b** $4 \times 5n =$ _____

**c** $6 \times 2a =$ _____ **d** $5 \times 8m =$ _____

**e** $9 \times 3k =$ _____ **f** $5y \times 4 =$ _____

**g** $7a \times 4 =$ _____ **h** $k \times 3 =$ _____

**i** $4m \times 1 =$ _____ **j** $4m \times 3n =$ _____

**k** $7a \times 4b =$ _____ **l** $8a \times 2b =$ _____

**m** $4k \times 5y =$ _____ **n** $5k \times 9m =$ _____

**o** $6p \times 7q =$ _____ **p** $3 \times 4a \times 5 =$ _____

**q** $7m \times n \times 5 =$ _____ **r** $2y \times k \times 6 =$ _____

**s** $4a \times 2b \times 3c =$ _____ **t** $4 \times 2b \times 7a =$ _____

**u** $6n \times 5k \times 2m =$ _____ **v** $3b \times a \times 5c =$ _____

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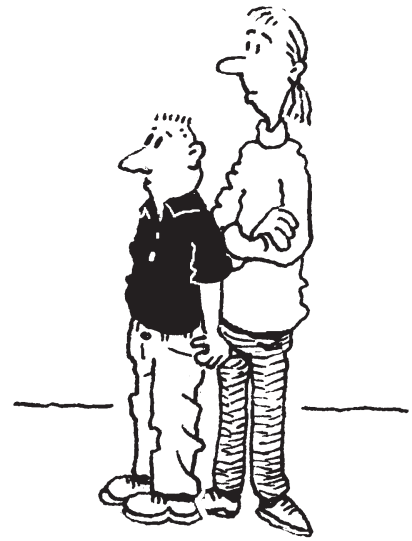
# 5.5 Multiplying with pronumerals

**CONTINUED**

Is there another way to write $a \times a$?

**Index notation** is a short way to write the number of times a pronumeral is multiplied by itself.

$a \times a$ becomes $a^2$.



**EXAMPLES**

$h \times h \times h \times h$
$= h^4$

$a \times a \times b \times b \times a$
$= a \times a \times a \times b \times b$
$= a^3b^2$

**4** Write using index notation.

**a** $n \times n \times n \times n =$ _____

**b** $a \times a \times a =$ _____

**c** $m \times m \times m \times m \times m =$ _____

**d** $y \times y =$ _____

**e** $a \times a \times b \times b \times b =$ _____

**f** $k \times m \times k \times m =$ _____

**g** $p \times q \times q =$ _____

**h** $a \times b \times a =$ _____

**i** $b \times c \times c \times b =$ _____

**j** $k \times k \times y \times y \times k =$ _____

**k** $a \times b \times a \times a \times b =$ _____

**l** $k \times n \times n \times k \times n =$ _____

**5** Simplify the following expressions.

**a** $9 \times m \times m \times m =$ _____

**b** $5 \times a \times 4 \times a =$ _____

**c** $8a \times 2a =$ _____

**d** $7y \times 3y =$ _____

**e** $d \times 3d =$ _____

**f** $5m \times 2m \times m =$ _____

**g** $5 \times n \times 2n =$ _____

**h** $3a \times a \times 4a =$ _____

**i** $6m \times 2m \times 4m =$ _____

**j** $2k \times 4k \times 3 =$ _____

**k** $4p \times 2p \times p \times p =$ _____

**l** $8m \times 2 \times 5m =$ _____

**m** $3k \times k \times y =$ _____

**n** $5m \times a \times m =$ _____

**o** $3m \times 2n \times 4m =$ _____

**p** $a \times b \times 2a =$ _____

**q** $a \times ab \times b =$ _____

**r** $2k \times 3y \times 5k =$ _____

**s** $3ac \times 2ac =$ _____

**t** $4m \times 3n \times mn =$ _____

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# 5.6 Division with pronumerals

**EXAMPLE**

**Ilse is in a group of four students.**
**They each had the same number of counters.**
**They have placed all the counters in one bucket.**
**How many counters did each student have?**

If the total number of counters = $N$,
each student had $N \div 4$ (or $\frac{N}{4}$) counters.

**1** The following quantities have been shared equally.
Write an algebraic expression for the amount that each student received.

**a** $N$ lollies divided among 5 students _____

**b** $n$ oranges divided among 4 students _____

**c** $Y$ pencils divided among 6 students _____

**d** 12 cakes divided among $n$ students _____

**e** 30 counters divided among $x$ students _____

To simplify algebraic divisions,
divide the numbers first.
Then divide (or cancel) the pronumerals.
Writing the division as a fraction
can make this easier.

**EXAMPLES**

| $6n \div 2$ | $10a \div 5a$ |
|---|---|
| $= \frac{6n}{2}$ | $= \frac{10a}{5a}$ |
| $= 3n$ | $= 2$ |

**2** Simplify these divisions.

**a** $\frac{15a}{3} =$ _____ **b** $\frac{10x}{5} =$ _____ **c** $\frac{36m}{4} =$ _____

**d** $18k \div 2 =$ _____ **e** $32n \div 8 =$ _____ **f** $20ab \div 4 =$ _____

**g** $\frac{3a}{3} =$ _____ **h** $\frac{ab}{a} =$ _____ **i** $\frac{ad}{4d} =$ _____

**j** $\frac{4ac}{2a} =$ _____ **k** $\frac{2xy}{xy} =$ _____ **l** $12mn \div 3n =$ _____

**m** $\frac{5a}{10} =$ _____ **n** $\frac{3m}{12} =$ _____ **o** $6y \div 30 =$ _____

**p** $\frac{3f}{9f} =$ _____ **q** $\frac{21ac}{7c} =$ _____ **r** $5xy \div 10y =$ _____

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- Divisibility and factors
- Square numbers and square roots
- Building patterns
- Using pattern rules
- Describing number patterns
- Writing rules using symbols
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